

General Review

Note

- **Closed-book exam**
- You should bring **student card** and **calculator**

Key Point for final exam

Quiz 1, Quiz 2 , Exercise, examples from Lecture book and homework.

Chapter 1:

Draw stem-and-left displays

Draw Boxplots

1.2 Pictorial and Tabular Method in Descriptive Statistics

■ Stem-and-Leaf Displays

Suppose we have a numerical data set $x_1, x_2, x_3, \dots, x_n$ for which each x_i consists of **at least two digits**.

Steps for constructing a Stem-and-Leaf Display

1. Select **one or more leading digits** for the **stem values**. The trailing digits become **the leaves**.
2. List possible **stem values** in a vertical column.
3. Record the leaf for every observation beside the corresponding stem value.
4. **Indicate the units** for stems and leaves someplace in the display.

1.2 Pictorial and Tabular Method in Descriptive Statistics

■ Example 2:

Observations: 16, 33, 64, 37, 31

Stem-and-Leaf Display

Stem		Leaf
1		6
3		3 7 1 [or 3 1 3 7]
6		4

Stem: tens digit

Leaf: ones digit

1.2 Pictorial and Tabular Method in Descriptive Statistics

■ Exercise 1:

Draw the stem-and-leaf display for data:

18, 8, 10, 43, 5, 30, 10, 22, 6, 27, 25, 58, 14, 18, 30, 41;

Stem		Leaf
0		5 6 8
1		0 0 4 8 8
2		2 5 7
3		0 0
4		1 3
5		8

Stem: tens digit

Leaf: ones digit

1.4 Measures of Variability

■ **Boxplots**

Describe several of a data set's **most prominent features**:

- center;
- spread;
- extent and nature of any departure from **symmetry** ;
- identification of “**outliers**”, **observations that lie unusually far from the main body of the data.**

1.4 Measures of Variability

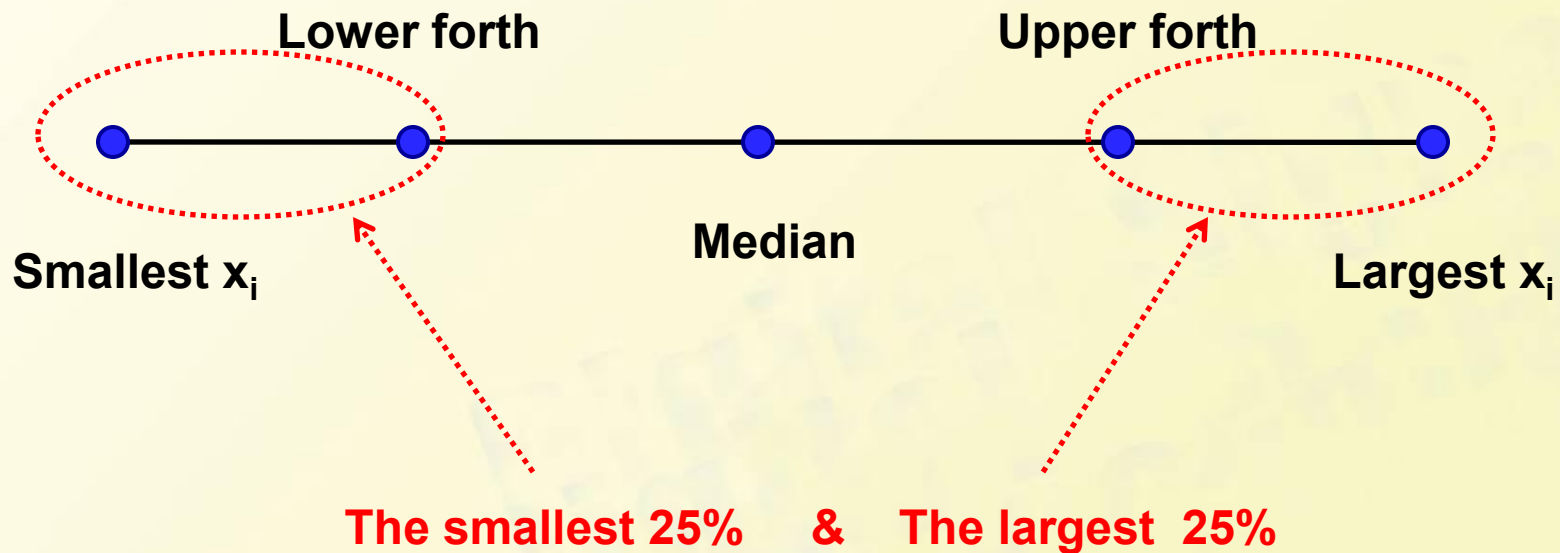
■ Fourth Spread

Order the n observations from smallest to largest and separate the smallest half from the largest half; the median is included in both halves if n is odd. Then the lower fourth is the median of the smallest half and the upper fourth is the median of the largest half. A measure of spread that is resistant to outliers is the fourth spread f_s , given by

$$f_s = \text{upper fourth} - \text{lower fourth}$$

1.4 Measures of Variability

- The simplest boxplot is based on the **5-number summary**



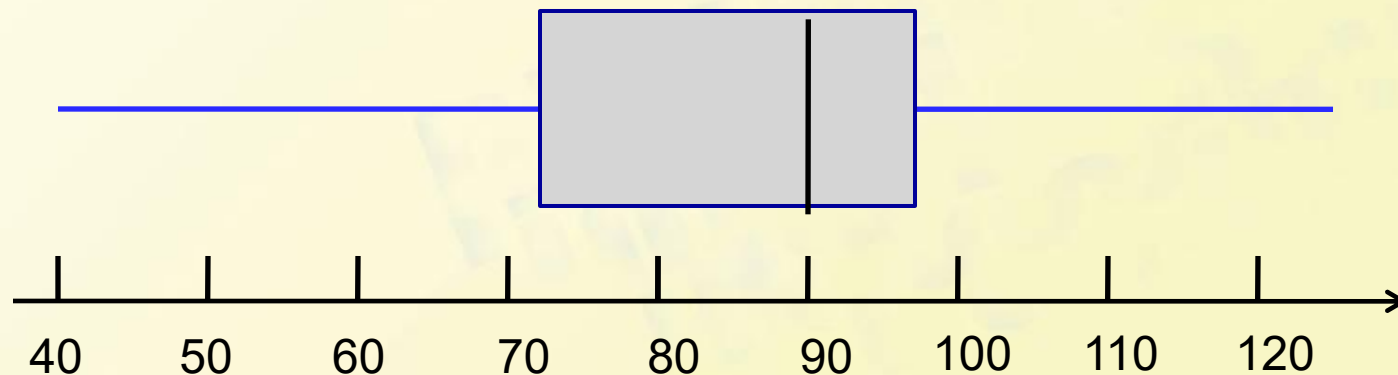
1.4 Measures of Variability

■ Example 1.19

40 52 55 60 70 75 85 85 90 90 92 94 94 95 98 100 115 125 125

Smallest x_i : 40 lower fourth = 72.5 median = 90

upper fourth = 96.5 largest x_i : 125



1.4 Measures of Variability

- A boxplot can be embellished to indicate explicitly the presence of **outliers**.
- **Outlier**: Any observation farther than $1.5 f_s$ from the closest fourth is an outlier.
- **Extreme**: An outlier is extreme if it is more than $3 f_s$ from the nearest fourth
- **Mild**: An outlier is mild if it is in the range of $(1.5 f_s, 3 f_s]$ from the nearest fourth.

1.4 Measures of Variability

■ Example

5.3 8.2 13.8 74.1 85.3 88.0 90.2 91.5 92.4
92.9 93.6 94.3 94.8 94.9 95.5 95.8 95.9 96.6
96.7 98.1 99.0 101.4 103.7 106.0 113.5

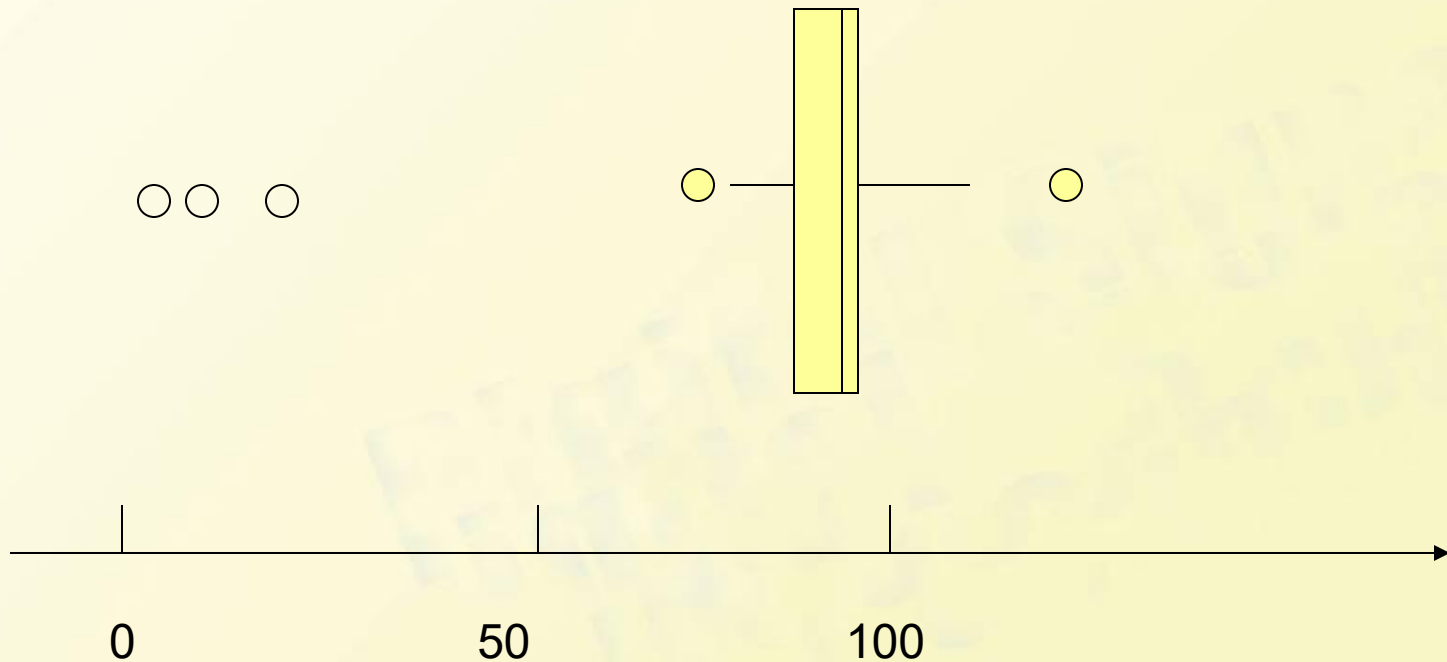
Relevant quantities

median = 94.8 lower fourth = 90.2 upper fourth = 96.7

$f_s = 6.5$ $1.5f_s = 9.75$ $3f_s = 19.5$

1.4 Measures of Variability

- A boxplot of the pulse width data showing mild and extreme outliers



Chapter 2:

- Equally Likely Outcomes
- Conditional Probability
- The Multiplication rule for $p(A \cap B)$
- Independence

• Equally Likely Outcomes

In many experiments consisting of N outcomes, it is reasonable to assign **equal probabilities to all N simple events**.

With $p=P(E_i)$ for every i , then

$$1 = \sum_{i=1}^N P(E_i) = \sum_{i=1}^N p = P \cdot N \quad \Rightarrow \quad p = 1/N$$

Consider an event A , with $N(A)$ denoting the number of outcomes containing in A , then

$$P(A) = \sum_{E_i \text{ in } A} P(E_i) = \sum_{E_i \text{ in } A} \frac{1}{N} = \frac{N(A)}{N}$$

Equally Likely Outcomes

- **Example 2.16**

When two dice are rolled separately, there are $N=36$ outcomes. If both the dice are fair, all 36 outcomes are equally likely.

- (A) What's the probability of elementary events?

(B) Let A be the event that sum of two number equal to 7, what's $P(A)$?

Solution:

(A) $P(E_i) = 1/36.$

(B) The event $A = \{\text{sum of two number} = 7\}$ consists of the six outcomes (1,6) (2,5) (3,4), (4,3), (5,2) and (6,1),

So

$$P(A) = N(A)/N = 6/36 = 1/6$$

2.2 Axioms, Interpretations, and Properties of Probability

- **Example 2.15**

Denote the **six elementary events** $\{1\}, \{2\}, \dots, \{6\}$ associated with **tossing a six-sided die once** by $E_1, E_2, \dots, E_6$.

If the die is constructed so that any of the **three even outcomes is twice** as likely to occur as any of the **three odd outcomes** (**unfair die**).

- (A) What's the probability of elementary events?
- (B) Let A be the event that outcome is even, what's $P(A)$?
- (C) Let B be the event that outcome ≤ 3 , what's $P(B)$?

Solution:

(A) The probabilities of **elementary events** is

$$P(E_1)=P(E_3)=P(E_5) = 1/9 \text{ and } P(E_2)=P(E_4)=P(E_6)=2/9,$$

(B) The event $A=\{\text{outcome is even}\} = E_2 \cup E_4 \cup E_6$

$$P(A) = P(E_2)+P(E_4)+P(E_6)=2/3$$

(C) The event $B=\{\text{outcome} \leq 3\} = E_1 \cup E_2 \cup E_3$

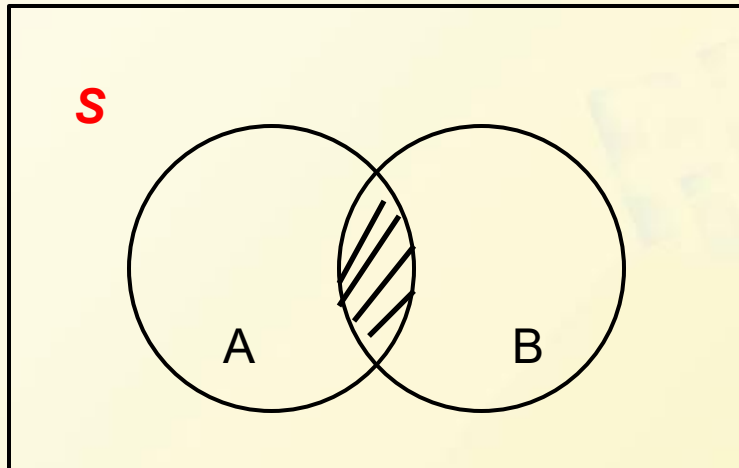
$$P(B) = P(E_1)+P(E_2)+P(E_3)=4/9$$

2.4 Conditional Probability

- Definition of Conditional Probability

For any two events A and B with $P(B) > 0$, the conditional probability of A given that B has occurred is defined by

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$



Note:

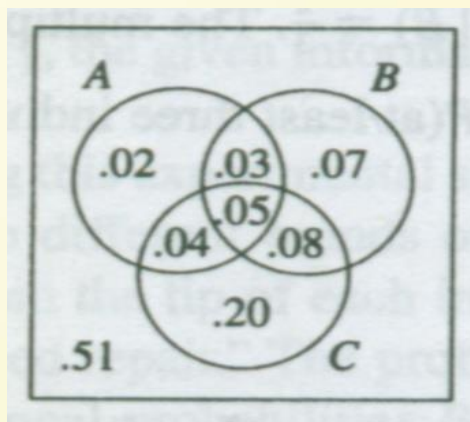
1. Given that B has occurred, the relevant sample space is **no longer S** but consists of **outcomes in B**;
2. A has occurred **if and only if** one of the outcomes in the **intersection** occurred.

2.4 Conditional Probability

- Example 2.26

A news magazine publishes three **columns** entitled “Art”(A), “Books”(B), and “Cinema”(C). Reading habits of a randomly selected reader with respect to these columns are

Read regularly	A	B	C	$A \cap B$	$A \cap C$	$B \cap C$	$A \cap B \cap C$
Probability	0.14	0.23	0.37	0.08	0.09	0.13	0.05



$$P(A|B) = 0.348$$

$$P(A|B \cup C) = 0.255$$

$$P(A|A \cup B \cup C) = 0.286$$

$$P(A \cup B|C) = 0.459$$

2.4 Conditional Probability

- The Multiplication Rule

$$P(A \cap B) = P(A | B) P(B)$$

This rule is important because it is often the case that $P(A \cap B)$ is desired, whereas that both $P(B)$ and $P(A|B)$ can be specified from the problem description.

Note :

1. $P(A \cap B) = P(A | B) P(B) = P(B | A) P(A)$
2. $P(A_1 \cap A_2 \cap A_3) = P(A_3 | A_1 \cap A_2) P(A_1 \cap A_2)$
 $= P(A_3 | A_1 \cap A_2) P(A_2 | A_1) P(A_1)$
3. $P(A_1 \cap A_2 \dots \cap A_n) = P(A_n | A_1 \cap A_2 \dots A_{n-1}) P(A_{n-1} | A_1 \cap A_2 \dots A_{n-2}) \dots P(A_2 | A_1) P(A_1)$

2.4 Conditional Probability

- **Example 2.27**

Four individuals have responded to request by a blood bank for blood donations. None of them has donated before, so their blood types are unknown. Suppose only type O+ is desired and only one of the four actually has this type. If the potential donors are selected in random order for typing, what is the probability that at least three individuals must be typed to obtain the desired type?

Solution:

Let $B = \{\text{first type not O+}\}$, $A = \{\text{second type not O+}\}$, so $P(B) = 3/4$

Given that the first type is not O+, two of the three individuals left are not O+, so $P(A|B) = 2/3$

The multiplication rule now gives:

$$\begin{aligned} P(\text{at least three individuals must be typed}) &= P(A \cap B) \\ &= P(A | B) P(B) \\ &= (2/3) * (3/4) \\ &= 0.5 \end{aligned}$$

• Example 2.29

A chain of video stores sells **three different brands** of VCRs. Of its VCR sales, 50% are brand 1, 30% are brand 2, and 20% are brand 3. Each manufacturer offers a 1-year warranty on parts and labor. It is known that **25% of brand 1's VCRs require warranty repair work**, whereas the corresponding percentages for **brands 2 and 3 are 20% and 10%, respectively**.

1. What is the probability that a randomly selected purchaser has bought **a brand 1 VCR that will need repair** while under warranty?
2. What is the probability that a randomly selected purchaser has **a VCR that will need repair** while under warranty?
3. If a customer returns to the store with a VCR that needs warranty repair work, what is the probability that it is a brand 1 VCR? A brand 2 VCR? A brand 3 VCR?

2.4 Conditional Probability

- **Solution:**

First Stage:

a customer selecting one of the three brands of VCR

Let $P(A_i) = \{\text{brand } i \text{ is purchased}\}$, where $i = 1, 2, 3$

then $P(A_1) = 0.5$, $P(A_2) = 0.3$, $P(A_3) = 0.2$

Second Stage:

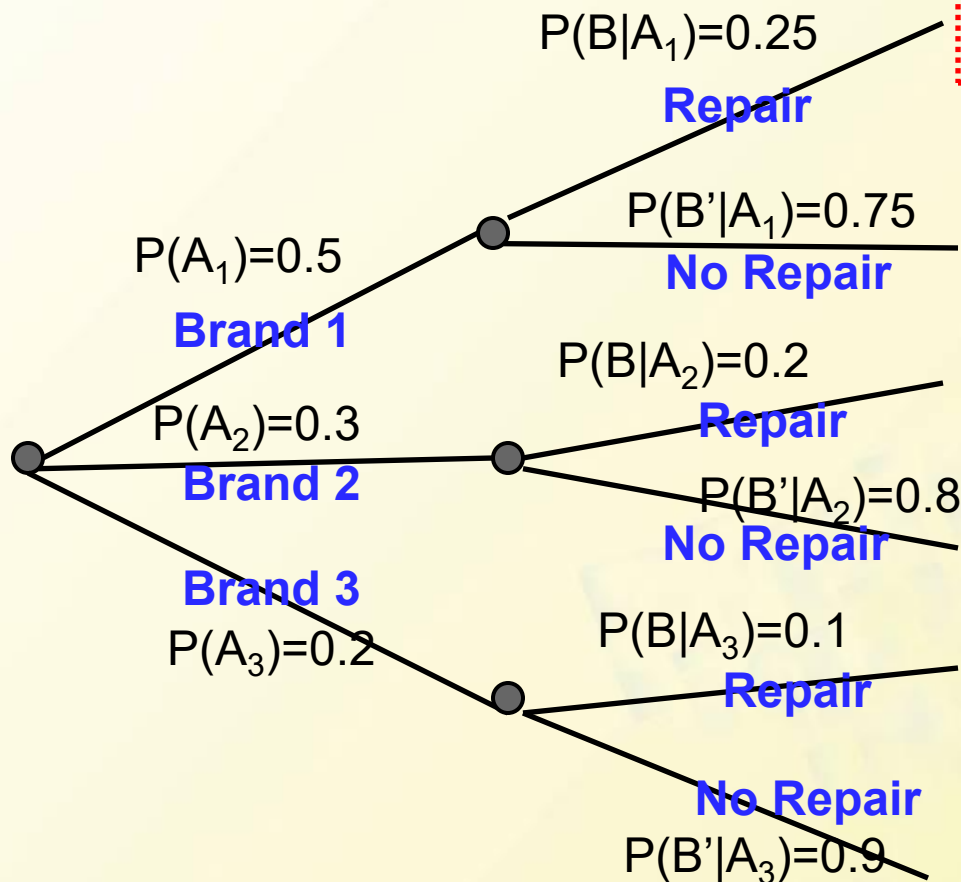
observing whether the selected VCR needs warranty repair

Let $B = \{\text{needs repair}\}$ $B' = \{\text{doesn't need repair}\}$

then $P(B|A_1) = 0.25$, $P(B|A_2) = 0.20$, $P(B|A_3) = 0.10$

2.4 Conditional Probability

- Example 2.29 (Cont')



Q1

$$P(A_1 \cap B) = P(A_1) P(B|A_1) = 0.125$$

$$P(A_2 \cap B) = P(A_2) P(B|A_2) = 0.06$$

$$P(A_3 \cap B) = P(A_3) P(B|A_3) = 0.02$$

Q2

$$P(B) = 0.125 + 0.06 + 0.02 = 0.205$$

2.4 Conditional Probability

- Example 2.29 (Cont')

Q3

$$P(A_1 | B) = \frac{P(A_1 \cap B)}{P(B)} = \frac{0.125}{0.205} = 0.61$$

$$P(A_2 | B) = \frac{P(A_2 \cap B)}{P(B)} = \frac{0.06}{0.205} = 0.29$$

$$P(A_3 | B) = 1 - P(A_2 | B) - P(A_1 | B) = 0.1$$

2.5 Independence

- **Definition**

Two events A and B are independence **if $P(A | B)=P(A)$** and are dependent otherwise.

Note:

1. Since $P(A \cap B) = P(A | B)P(B) = P(B | A)P(A)$

if $P(A | B)=P(A)$, then we have

$$P(A) P(B) = P(B|A)P(A) \rightarrow P(B|A) = P(B) \text{ (if } P(A) > 0\text{)}$$

2. If A and B are independence, so are the **following pairs of events:**

a. A' and B b. A and B' c. A' and B'

2.5 Independence

- **Proposition #1**

A and B are independent *if and only if*

$$P(A \cap B) = P(A) P(B)$$

Proof:

1. If A and B are independent, then

$$P(A|B) = P(A), \text{ and thus}$$

$$P(A \cap B) = P(A|B)P(B) = P(A) P(B)$$

2. If $P(A \cap B) = P(A) P(B)$, then

$$P(A \cap B) = P(A|B)P(B) = P(A) P(B)$$

$$P(A|B) = P(A) \quad (P(B) > 0), \text{ A and B are independent}$$

2.5 Independence

Example 2.35

Consider first the system illustrated in Figure 2.14(a).

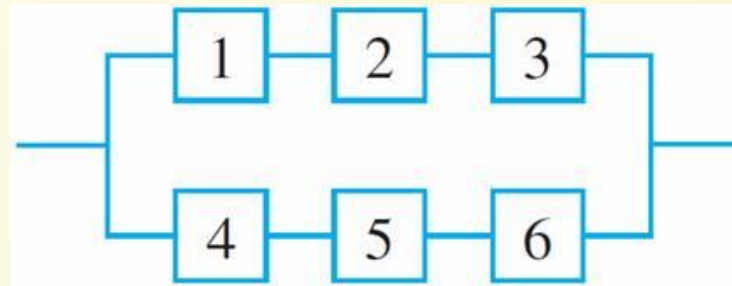


Figure 2.14(a)

There are two **subsystems connected in parallel**, each one containing three cells. In order for the system to function, **at least one of the two parallel subsystems must work**.

Within each subsystem, the three cells are **connected in series**, so a subsystem will work **only if all cells in the subsystem work**.

- Let A_i denote the event that the lifetime of cell i exceeds a particular lifetime value t_0 ($i = 1, 2, \dots, 6$).
- We assume that the A_i 's are independent events (whether any particular cell lasts more than t_0 hours has no bearing on whether or not any other cell does) and that $P(A_i) = .9$ for every i since the cells are identical.
- What is the probability that the system lifetime exceeds t_0 ?

Solution:

$P(\text{system lifetime exceeds } t_0)$

$$= P[(A1 \cap A2 \cap A3) \cup (A4 \cap A5 \cap A6)]$$

$$= P(A1 \cap A2 \cap A3) + P(A4 \cap A5 \cap A6) \\ - P[(A1 \cap A2 \cap A3) \cap (A4 \cap A5 \cap A6)]$$

$$= (.9)(.9)(.9) + (.9)(.9)(.9) - (.9)(.9)(.9)(.9)(.9)(.9)$$

$$= .927$$

Alternatively,

P (system lifetime exceeds t_0)

$$= 1 - P(\text{both subsystem lives are } \leq t_0)$$

$$= 1 - [P(\text{subsystem life is } \leq t_0)]^2$$

$$= 1 - [1 - P(\text{subsystem life is } > t_0)]^2$$

$$= 1 - [1 - (.9)^3]^2$$

$$= .927$$

Next consider the **total-cross-tied system** shown in **Figure 2.14(b)**, obtained from the **series-parallel array** by connecting ties across each column of junctions.

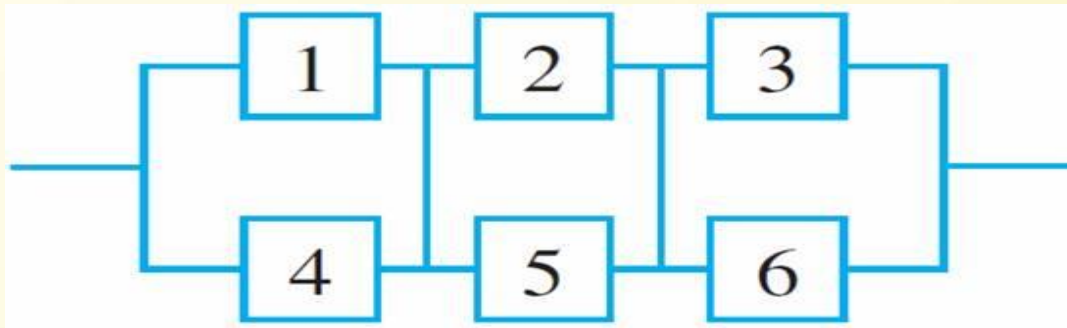


Figure 2.14(b)

Now the system **fails as soon as an entire column fails**, and system lifetime exceeds t_0 only if the life of every column does so.

For this configuration,

P (system lifetime is at least t_0)

$$= [P (\text{column lifetime exceeds } t_0)]^3$$

$$= [1 - P (\text{column lifetime} \leq t_0)]^3$$

$$= [1 - P (\text{both cells in a column have lifetime} \leq t_0)]^3$$

$$= [1 - (1 - .9)^2]^3$$

$$= .970$$

Chapter 3:

- Probability Distribution
- Cumulative Distribution Function
- Expected Value
- The Variance
- The Binomial Probability Distribution
- The Poisson Probability Distribution

3.2 Probability Distributions for Discrete Random Variables

■ Probability Distribution

The **probability distribution** or **probability mass function** (pmf) of a discrete rv is defined for every number x by

$$p(x) = P(X=x) = P(\text{all } s \text{ in } S: X(s)=x)$$

Note:

$$p_i = p(x_i) \geq 0, \sum_{i=1}^n p_i = 1$$

x	x_1	x_2	$\dots$	x_n
$p(x)$	p_1	p_2	$\dots$	p_n

Probability Distribution

■ Example 3.10

Consider a group of **five potential** blood donors—A,B,C,D, and E—of whom only **A and B have type O+ blood**. Five blood samples, one from each individual, **will be typed in random order until an O+ individual is identified**.

Let the rv Y = the number of typings necessary to identify an O+ individual.

Then the pmf of Y is

y	1	2	3	4
$p(y)$	0.4	0.3	0.2	0.1

Where any y value not listed relieves zero probability

3.2 Probability Distributions for Discrete Random Variables

■ Cumulative Distribution Function

The cumulative distribution function (cdf) $F(x)$ of a discrete rv variable X with pmf $p(x)$ is defined for every number x by

$$F(x) = P(X \leq x) = \sum_{y: y \leq x} p(y)$$

For any number x , $F(x)$ is the probability that the observed value of X will be at most x .

3.2 Probability Distributions for Discrete Random Variables

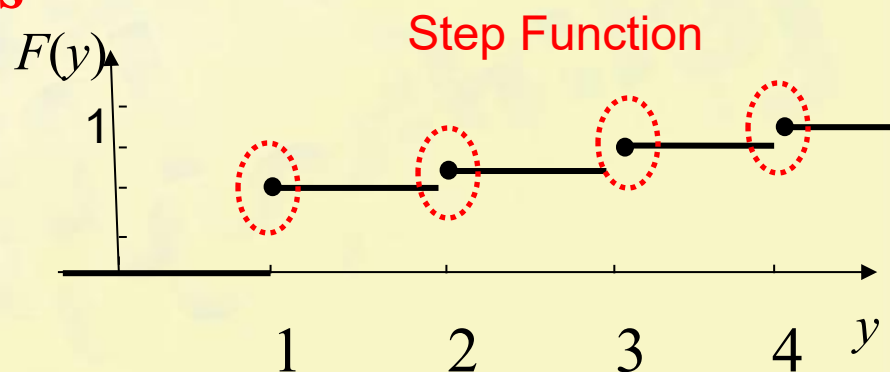
■ Example (Ex. 3.10 continued)

The pmf of Y (the number of blood typings) in Example 3.10 was

y	1	2	3	4
$p(y)$	0.4	0.3	0.2	0.1

Then the corresponding cdf is

$$F(y) = \begin{cases} 0, & \text{if } y < 1 \\ 0.4 & \text{if } 1 \leq y < 2 \\ 0.7 & \text{if } 2 \leq y < 3 \\ 0.9 & \text{if } 3 \leq y < 4 \\ 1, & \text{if } 4 \leq y \end{cases}$$



3.2 Probability Distributions for Discrete Random Variables

■ Example 3.14 (Ex. 3.12 Cont')

$$p(x) = \begin{cases} (1-p)^{x-1} p, & x = 1, 2, 3, \dots \\ 0 & \text{otherwise} \end{cases}$$

For a positive integer x ,

$$F(x) = \sum_{y \leq x} p(y) = \sum_{y=1}^x (1-p)^{y-1} p = p \sum_{y=1}^x (1-p)^{y-1}$$

$$F(x) = p \cdot \frac{1 - (1-p)^x}{1 - (1-p)} = 1 - (1-p)^x$$

For any real value x ,

$$F(x) = 1 - (1-p)^{\lfloor x \rfloor} \quad \lfloor x \rfloor \text{ is the largest integer } \leq x$$

3.2 Probability Distributions for Discrete Random Variables

■ Proposition

For any two numbers a and b with $a \leq b$,

$$P(a \leq X \leq b) = F(b) - F(a-)$$

where “ $a-$ ” represents the largest possible X value that is strictly less than a .

In particular, if the only possible values are integers and if a and b are integers, then

$$\begin{aligned} P(a \leq X \leq b) &= P(X=a \text{ or } a+1 \text{ or } \dots \text{ or } b) \\ &= F(b) - F(a-1) \end{aligned}$$

Taking $a=b$ yields $P(X = a) = F(a) - F(a-1)$ in this case.

3.2 Probability Distributions for Discrete Random Variables

■ Example 3.15

Let X = the number of days of sick leave taken by a randomly selected employee of a large company during a particular year.

If the maximum number of allowable sick days per year is 14, possible values of X are 0, 1, ..., 14. With $F(0)=0.58$, $F(1)=0.72$, $F(2)=0.76$, $F(3)=0.81$, $F(4)=0.88$, and $F(5) = 0.94$. Then :

$$P(2 \leq X \leq 5) = ?$$

$$P(X=3) = ?$$

3.2 Probability Distributions for Discrete Random Variables

Solution:

$$P(2 \leq X \leq 5) = P(X=2,3,4 \text{ or } 5) = F(5) - F(1) = 0.22$$

$$P(X=3) = F(3) - F(2) = 0.05$$

3.2 Probability Distributions for Discrete Random Variables

■ Three Properties of cdf (discrete/continuous cases)

1. Non-decreasing, *i.e.* if $x_1 < x_2$ then $F(x_1) \leq F(x_2)$

2.
$$F(-\infty) = \lim_{x \rightarrow -\infty} F(x) = 0$$

$$F(+\infty) = \lim_{x \rightarrow +\infty} F(x) = 1$$

3. $F(x+0) = F(x)$

Note: Any function that satisfies the above properties would be a cdf.

3.3 Expected Values of Discrete Random Variables

■ The Expected Value of X

Let X be a discrete rv with set of possible values D and pmf $p(x)$. The expected value or **mean value** of X , denoted by $E(X)$ or μ_X (or μ for short), is

$$E(X) = \mu_X = \sum_{x \in D} x \cdot p(x)$$

Note: When the sum does not exist, we say the expectation of X does not exist. (finite or infinite case?)

3.3 Expected Values of Discrete Random Variables

■ Example 3.19

The general form for the pmf of **X=number of children born up to and including the first boy** is

$$p(x) = \begin{cases} p(1-p)^{x-1} & x=1,2,3,\dots \\ 0 & \text{otherwise} \end{cases}$$

E(x)=?

3.3 Expected Values of Discrete Random Variables

■ Example 3.19

Solution:

$$\begin{aligned} E(x) &= \sum_D x \cdot p(x) = \sum_{x=1}^{\infty} xp(1-p)^{x-1} = p \sum_{x=1}^{\infty} \left[-\frac{d}{dp} (1-p)^x \right] \\ &= -p \frac{d}{dp} \sum_{x=1}^{\infty} (1-p)^x = -p \frac{d \left[\frac{1-p}{1-(1-p)} \right]}{dp} = -p \left[\frac{1-p}{p} \right]' = \frac{1}{p} \end{aligned}$$

3.3 Expected Values of Discrete Random Variables

- The Expected Value of a **function**

Let X be a discrete rv with set of **possible values** D and pmf $p(x)$. Then the expected values or mean value of any function $h(X)$, denoted by **$E[h(X)]$** or **$\mu_{h(X)}$** , is computed by

$$E[h(X)] = \sum_{x \in D} h(x) \cdot p(x)$$

3.3 Expected Values of Discrete Random Variables

■ Example 3.23

A computer store has purchased three computers of a certain type at \$500 apiece. It will sell them for \$1000 apiece. The manufacturer has agreed to repurchase any computers still unsold after specified period at \$200 apiece.

Let X denote the number of computers sold, and suppose that $p(0)=0.1$, $p(1)=0.2$, $p(2)=0.3$ and $p(3)=0.4$. The $h(x)$ denoting the profit associated with selling X units.

What is the profit and expected profit?

3.3 Expected Values of Discrete Random Variables

Solution:

$$h(X) = \text{revenue} - \text{cost} = 1000X + 200(3-X) - 1500 = 800X - 900.$$

The expected profit is

$$\begin{aligned} E(h(X)) &= h(0)p(0) + h(1)p(1) + h(2)p(2) + h(3)p(3) \\ &= (-900)(0.1) + (-100)(0.2) + (700)(0.3) + (1500)(0.4) \\ &= 700 \end{aligned}$$

3.3 Expected Values of Discrete Random Variables

■ Rule of Expected Value

$$E(aX+b) = a E(X) + b$$

Proof:

$$\begin{aligned} E(aX + b) &= \sum_D (ax + b)p(x) \\ &= a \sum_D xp(x) + b \sum_D p(x) \\ &= aE(x) + b \end{aligned}$$

1. For any constant a , $E(aX)=aE(X)$ ($b=0$)
2. For any constant b , $E(X+b)=E(X)+b$ ($a=1$)

3.3 Expected Values of Discrete Random Variables

■ The Variance of X

Let X have pmf $p(x)$ and the expected value μ . Then the variance of X, denoted by $V(X)$ or σ_x^2 , or just σ^2 , is

$$V(X) = \sum_{x \in D} (x - \mu)^2 \cdot p(x) = E[(X - \mu)^2]$$

The standard deviation (SD) of X is

$$\sigma_X = \sqrt{\sigma_X^2}$$

3.3 Expected Values of Discrete Random Variables

■ Example

If X is the number of cylinders on the next car to be tuned at a service facility, with pmf as given, then **what is the $V(X)$?**

x	4	6	8
$p(x)$	0.5	0.3	0.2

3.3 Expected Values of Discrete Random Variables

Solution:

$$\begin{aligned} V(X) = \sigma^2 &= \sum_{x=4}^8 (x - 5.4)^2 \cdot p(x) \\ &= (4 - 5.4)^2 (.5) + (6 - 5.4)^2 (.3) + (8 - 5.4)^2 (.2) = 2.44 \end{aligned}$$

3.3 Expected Values of Discrete Random Variables

■ A short formula for σ^2

$$V(X) = \sigma^2 = \left[\sum_D x^2 p(x) \right] - \mu^2 = E(X^2) - [E(X)]^2$$

Proof:

$$\begin{aligned} V(X) &= \sum_{x \in D} (x - \mu)^2 \cdot p(x) = E[(X - \mu)^2] \\ &= \sum_{x \in D} x^2 \cdot p(x) - 2\mu \sum_D xp(x) + \mu^2 \sum_D p(x) \\ &= E(X^2) - 2\mu\mu + \mu^2 = E(X^2) - \mu^2 \\ &= E(X^2) - [E(X)]^2 \end{aligned}$$

3.3 Expected Values of Discrete Random Variables

■ Example 3.25

The pmf of the number of cylinders X on the next car to be turned at a certain facility was given in Example 3.24 as $p(4)=0.5$, $p(6)=0.3$ and $p(8)=0.2$, from which $\mu=5.4$, and

$$E(X^2) = (4^2)(0.5) + (6^2)(0.3) + (8^2)(0.2) = 31.6$$

$$\sigma^2 = E(X^2) - E(X)^2 = 31.6 - (5.4)^2 = 2.44$$

3.3 Expected Values of Discrete Random Variables

■ Rules of Variance

$$V(aX + b) = \sigma_{aX+b}^2 = a^2 \sigma_X^2 \quad \& \quad \sigma_{aX+b} = |a| \sigma_X$$

$$1. \quad \sigma_{aX}^2 = a^2 \sigma_X^2, \sigma_{aX} = |a| \sigma_X$$

$$2. \quad \sigma_{X+b}^2 = \sigma_X^2$$

3.3 Expected Values of Discrete Random Variables

■ Example 3.26

In the computer sales problem of **Example 3.23**, $E(X)=2$ and

$$E(X^2)=(0)^2(0.1)+(1)^2(0.2)+(2)^2(0.3)+(3)^2(0.4)=5$$

so $V(X)=5-(2)^2=1$.

What is the variance and deviation of profit function $h(X)=800X-900$?

Solution:

**The variance: $(800)^2V(X)=(640,000)(1)=640,000$
standard deviation 800.**

3.4 The Binomial Probability Distribution

- The requirements for a **binomial experiment**
 1. The experiment consists of a sequence of n smaller experiments called trials, where n is fixed in advance of the experiment.
 2. Each trial can result in one of the same two possible outcomes (dichotomous trials), which we denote by success (S) or failure (F).
 3. The trials are independent, so that the outcome on any particular trial does not influence the outcome on any other trial.
 4. The probability of success is constant from trial to trial; we denote this probability by p .

3.4 The Binomial Probability Distribution

■ Binomial random variable

Definition:

Given a binomial experiment consisting of n trials, the binomial random variable X associated with this experiment is defined as

$X =$ the number of S's among the n trials

Suppose, for instance, that $n=3$. Then there are eight possible outcomes for the experiment:

SSS SSF SFS SFF FSS FSF FFS FFF

$X(\text{SSS}) = 3, X(\text{SSF}) = 2, \dots X(\text{FFF}) = 0$

3.4 The Binomial Probability Distribution

■ $X \sim \text{Bin}(n, p)$

Possible values for X in an n -trial experiment are $x = 0, 1, 2, \dots, n$. we will often write $X \sim \text{Bin}(n, p)$ to indicate that X is a binomial rv based on n trials with success probability p .

Notation:

Because the pmf of a binomial rv depends on the two parameters n and p , we denote the pmf by $b(x; n, p)$

3.4 The Binomial Probability Distribution

- Theorem:

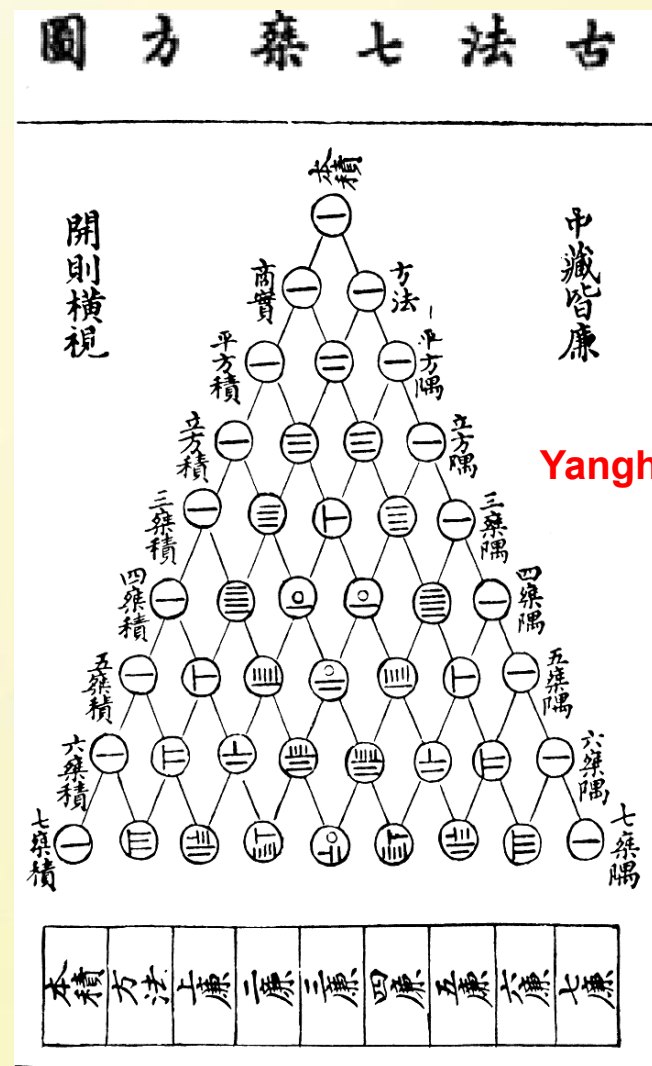
$$b(x; n, p) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x}, & x = 0, 1, 2, \dots, n \\ 0 & , \text{ otherwise} \end{cases}$$

3.4 The Binomial Probability Distribution

$$(a + b)^n = \sum_{x=0}^n \binom{n}{x} a^x b^{n-x}$$

$$\sum_{x=0}^n b(x; n, p) = \sum_{x=0}^n \binom{n}{x} p^x (1-p)^{n-x}$$

$$= [p + (1-p)]^n = 1$$



3.4 The Binomial Probability Distribution

■ Example 3.31

Each of six randomly selected cola drinkers is given a glass containing **cola S** and one containing **cola F**. The glasses are identical in appearance except for a code on the bottom to identify the cola. Suppose there is actually no tendency among cola drinkers to prefer one cola to the other. **Then $p = P(\text{a selected individual prefers S}) = 0.5$** , so with X = the number among the six who prefer S, $X \sim \text{Bin}(6, 0.5)$.

$$P(X = 3) = b(3; 6, 0.5) = \binom{6}{3} (0.5)^3 (0.5)^3 = 20(0.5)^6 = 0.313$$

$$P(3 \leq X) = \sum_{x=3}^6 b(x; 6, 0.5) = \sum_{x=3}^6 \binom{6}{x} (0.5)^x (0.5)^{6-x} = 0.656$$

3.4 The Binomial Probability Distribution

■ Notation

For $X \sim \text{Bin}(n, p)$, the **cdf** will be denoted by

$$P(X \leq x) = B(x; n, p) = \sum_{y=0}^x b(y; n, p), x = 0, 1, \dots, n$$

■ Binomial Table

Refer to Appendix Table A.1

3.4 The Binomial Probability Distribution

- $B(n,p)$ with $n=5$, $p=0.1, 0.3, 0.5, 0.7$ and 0.9

x	p					
		0.1	0.3	0.5	0.7	0.9
0		0.590	0.168	0.031	0.002	0.000
1		0.919	0.528	0.188	0.031	0.000
2		0.991	0.837	0.500	0.163	0.009
3		1.000	0.969	0.812	0.472	0.081
4		1.000	0.998	0.969	0.832	0.410
5		1.000	1.000	1.000	1.000	1.000

$B(3; 5, 0.5)$

$B(2; 5, 0.7)$

3.4 The Binomial Probability Distribution

■ Example 3.32

Suppose that 20% of all copies of a particular textbook fail a certain binding strength test. Let X denote the number among 15 randomly selected copies that fail the test. Then X has a binomial distribution with $n=15$ and $p=0.2$.

1. The probability that at most 8 fail the test is

$$P(X \leq 8) = \sum_{y=0}^8 b(y; 15, 0.2) = B(8; 15, 0.2) = 0.999$$

2. The probability that exactly 8 fail is

$$P(X = 8) = P(X \leq 8) - P(X \leq 7) = B(8; 15, 0.2) - B(7; 15, 0.2) = 0.999 - 0.996 = 0.003$$

3. The probability that at least 8 fail is

$$P(X \geq 8) = 1 - P(X \leq 7) = 1 - B(7; 15, 0.2) = 1 - 0.996 = 0.004$$

4. The probability that between 4 and 7

$$P(4 \leq X \leq 7) = P(X \leq 7) - P(X \leq 3) = B(7; 15, 0.2) - B(3; 15, 0.2) = 0.996 - 0.648 = 0.348$$

3.4 The Binomial Probability Distribution

■ The Mean and Variance of X

Since a **binomial experiment** consists of n trials, intuition suggests that for

$$X \sim \text{Bin}(n, p) \quad , \quad E(X) = np$$

But the expression for $V(X)$ is not so intuitive

3.4 The Binomial Probability Distribution

■ Proposition

If $X \sim \text{Bin}(n, p)$, then $E(X) = np$, $V(X) = np(1-p) = npq$, and

$$\sigma_X = \sqrt{npq}$$

where $q = 1-p$

3.6 The Poisson Probability Distribution

The **Poisson** distribution provides a useful model for many random phenomena.

It can be used as **a limiting form** of the binomial distribution

3.6 The Poisson Probability Distribution

■ Poisson Distribution

A random variable X is said to have a **Poisson distribution** with parameter λ ($\lambda > 0$) if the **pmf** of X is

$$p(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!} \quad x = 0, 1, 2, 3, \dots$$

The value of λ is frequency a rate per unit time or per unit area. The constant e is the base of the natural logarithm system.

3.6 The Poisson Probability Distribution

- The **Maclaurin** infinite series expansion of e^λ

$$e^\lambda = 1 + \lambda + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} + \dots = \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} \quad (3.19)$$

if the two extreme terms in Expression (3.19) are multiplied by $e^{-\lambda}$ and then $e^{-\lambda}$ is placed inside the summation, the result is

$$1 = \sum_{n=0}^{\infty} e^{-\lambda} \frac{\lambda^n}{n!}$$

Which shows that $p(x; \lambda)$ fulfills the second condition necessary for specifying a pmf

3.6 The Poisson Probability Distribution

■ Proposition

If X has a Poisson distribution with parameter λ , then

$$E(X) = V(X) = \lambda.$$

Proof:

$$E(X) = \sum_{x=0}^{\infty} x \frac{e^{-\lambda} \lambda^x}{x!} = \sum_{x=1}^{\infty} \frac{e^{-\lambda} \lambda^x}{(x-1)!} = \sum_{y=0}^{\infty} \frac{e^{-\lambda} \lambda^{y+1}}{y!} = \lambda e^{-\lambda} \sum_{y=0}^{\infty} \frac{\lambda^y}{y!} = \lambda$$

$$\begin{aligned} E(X^2) &= \sum_{x=0}^{\infty} x^2 \frac{e^{-\lambda} \lambda^x}{x!} = \lambda e^{-\lambda} \sum_{x=1}^{\infty} x \frac{\lambda^{x-1}}{(x-1)!} \\ &= \lambda e^{-\lambda} \left\{ \sum_{x=1}^{\infty} [(x-1) \frac{\lambda^{x-1}}{(x-1)!}] + \left[\frac{\lambda^{x-1}}{(x-1)!} \right] \right\} = \lambda e^{-\lambda} \left[\lambda \sum_{x=2}^{\infty} \frac{\lambda^{x-2}}{(x-2)!} + \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!} \right] \\ &= \lambda e^{-\lambda} [\lambda e^{\lambda} + e^{\lambda}] = \lambda^2 + \lambda \end{aligned}$$

$$V(X) = E(X^2) - E(X)^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$

3.6 The Poisson Probability Distribution

■ Example 3.39

Let X denote the number of creatures of a particular type captured in a trap during a given time period. **Suppose that X has a Poisson distribution with $\lambda=4.5$** , so on average traps will contain 4.5 creatures. The probability that a trap contains **exactly five creatures** is

$$P(X = 5) = \frac{e^{-4.5} (4.5)^5}{5!} = 0.1708$$

The probability that a trap has **at most five creatures** is

$$P(X \leq 5) = \sum_{x=0}^5 \frac{e^{-4.5} (4.5)^x}{x!} = e^{-4.5} \left[1 + 4.5 + \frac{(4.5)^2}{2!} + \dots + \frac{(4.5)^5}{5!} \right] = 0.7029$$

3.6 The Poisson Probability Distribution

■ Example (Ex. 3.39 Cont')

Both the expected number of creatures trapped and the variance of the number trapped equal 4.5, and

$$\delta_x = (4.5)^{1/2} = 2.12$$

3.6 The Poisson Probability Distribution

■ The Poisson Distribution as a Limit

The rationale for using the Poisson distribution in many situations is provided by the following proposition.

Proposition:

Suppose that in the **binomial pmf** $b(x;n,p)$, we let $n \rightarrow \infty$ and $p \rightarrow 0$ in such a way that np approaches a value $\lambda > 0$. Then $b(x;n,p) \rightarrow p(x; \lambda)$

According to this proposition, in any binomial experiment in which n is large and p is small, $b(x;n,p) \approx p(x; \lambda)$

As a rule, this approximation can safely be applied if $n \geq 100$, and $p \leq 0.01$ and $np \leq 20$

3.6 The Poisson Probability Distribution

■ Example 3.40

If a publisher of nontechnical books takes great pains to ensure that its books are free of **typographical** errors, so that the probability of any given page containing **at least one such error is 0.005** and errors are independent from page to page, what is the probability that one of its **400-page novels will contain exactly one page with errors?**
At most three pages with errors?

3.6 The Poisson Probability Distribution

Solution:

With S denoting a page containing at least one error and F an error-free page, the number X of pages containing at least one error is a binomial rv with $n = 400$ and $p = 0.005$, so $np=2$. We wish

$$P(X = 1) = b(1; 400, 0.005) \approx p(1; 2) = \frac{e^{-2}(2)^1}{1!} = 0.271$$

$$\&P(X \leq 3) \approx \sum_{x=0}^3 p(x; 2) = \sum_{x=0}^3 e^{-2} \frac{2^x}{x!} = 0.135 + 0.271 + 0.271 + 0.180 = 0.857$$

Chapter 4:

- Probability Distribution for Continuous RV
- Uniform Distribution
- Cumulative Distribution Function
- Expected Value and variance
- Normal distribution

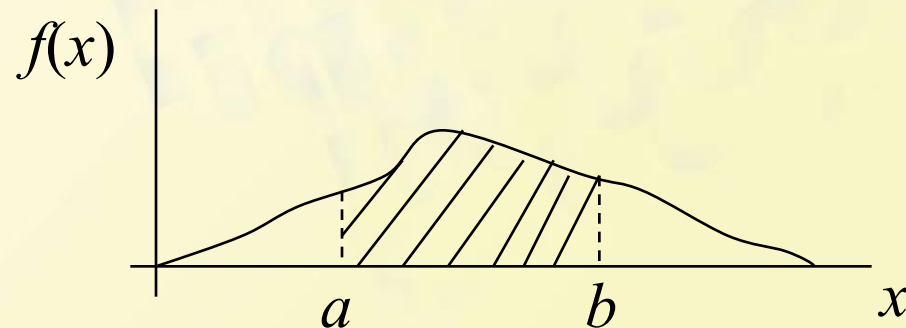
4.1 Continuous Random Variables and Probability Density Functions

■ Probability Distribution

Let X be a continuous rv. Then a **probability distribution** or **probability density function (pdf)** of X is $f(x)$ such that for any two numbers a and b with $a \leq b$

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

The probability that X takes on a value in the interval $[a,b]$ is the **area** under the graph of the density function as follows.

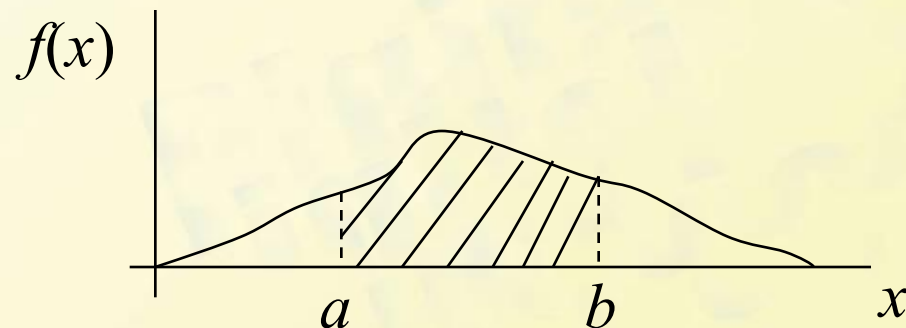


4.1 Continuous Random Variables and Probability Density Functions

- A legitimate pdf should satisfy

1. $f(x) \geq 0$ for all x

2. $\int_{-\infty}^{\infty} f(x)dx = \text{area under the entire graph of } f(x)$
 $= 1$



4.1 Continuous Random Variables and Probability Density Functions

■ Uniform Distribution

A continuous rv X is said to have a **uniform distribution** on the interval $[A, B]$ if the **pdf** of X is

$$f(x; A, B) = \begin{cases} \frac{1}{B - A} & A \leq x \leq B \\ 0 & \text{otherwise} \end{cases}$$

4.1 Continuous Random Variables and Probability Density Functions

■ Example 4.5

“Time headway” in traffic flow is the **elapsed time** between the time that one car finishes passing a fixed point and the instant that the next car begins to pass that point.

Let X = the time headway for two randomly chosen consecutive cars on a freeway during a period of heavy flow.

The following pdf of X is essentially the one suggested in “The Statistical Properties of Freeway Traffic”.

$$f(x) = \begin{cases} 0.15e^{-0.15(x-0.5)} & x \geq 0.5 \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

(A) The formula (1) satisfy the pdf condition?

(B) The probability that headway time is at most 5 sec is ?

4.1 Continuous Random Variables and Probability Density Functions

■ Solution:

$$f(x) = \begin{cases} 0.15e^{-0.15(x-0.5)} & x \geq 0.5 \\ 0 & \text{otherwise} \end{cases}$$

1. $f(x) \geq 0$;

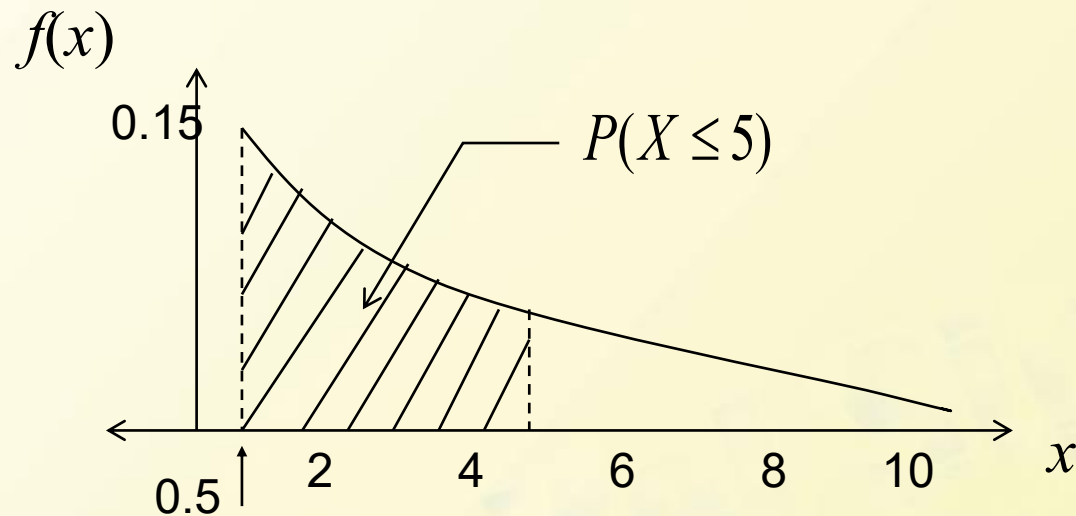
2. To show $\int_{-\infty}^{\infty} f(x) dx = 1$, we use the result $\int_a^{\infty} e^{-kx} dx = \frac{1}{k} e^{-ka}$

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) dx &= \int_{0.5}^{\infty} 0.15e^{-0.15(x-0.5)} dx = 0.15e^{0.075} \int_{0.5}^{\infty} e^{-0.15x} dx \\ &= 0.15e^{0.075} \cdot \frac{1}{0.15} e^{-(0.15)(0.5)} = 1 \end{aligned}$$

Thus, the formula (1) satisfy the pdf condition.

4.1 Continuous Random Variables and Probability Density Functions

■ Example 4.5 (Cont')



$$\begin{aligned} P(X \leq 5) &= \int_{-\infty}^5 f(x) dx = \int_{0.5}^5 .15e^{-0.15(x-5)} dx = 0.15e^{0.075} \int_{0.5}^5 e^{-0.15x} dx \\ &= 0.15e^{0.075} \cdot \left(-\frac{1}{0.15} e^{-0.15x} \right) \bigg|_{0.5}^5 = 0.491 = P(X < 5) \end{aligned}$$

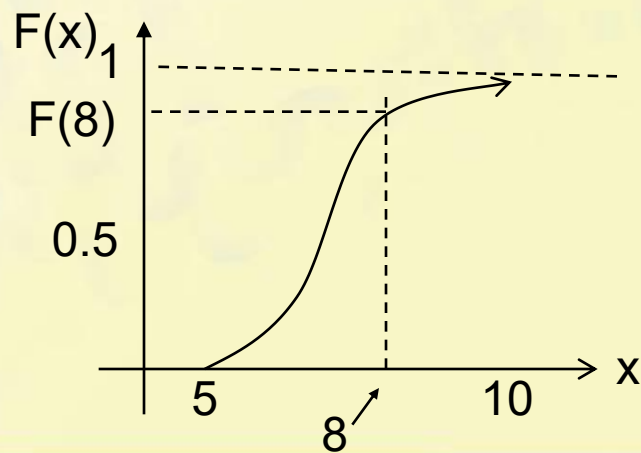
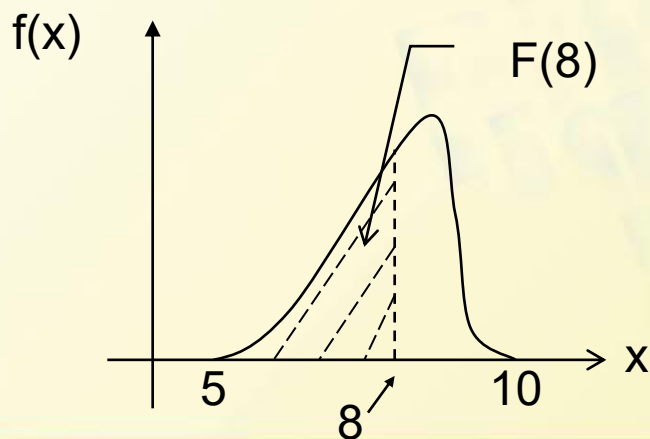
4.2 Cumulative Distribution Functions and Expected Values

■ Cumulative Distribution Function

The cumulative distribution function $F(x)$ for a continuous rv X is defined for every number x by

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(y)dy$$

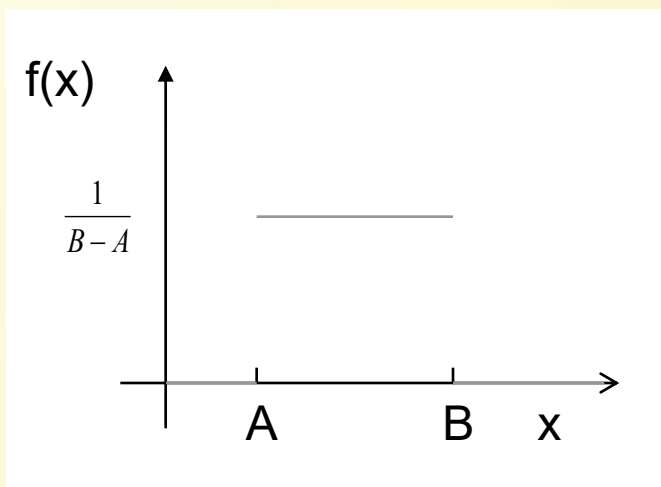
For each x , $F(x)$ is the **area** under the **density curve** to the **left of x** as follows



4.2 Cumulative Distribution Functions and Expected Values

■ Example 4.6

Let X , the thickness of a certain metal sheet, have a **uniform distribution** on $[A, B]$. The density function is shown as follows.

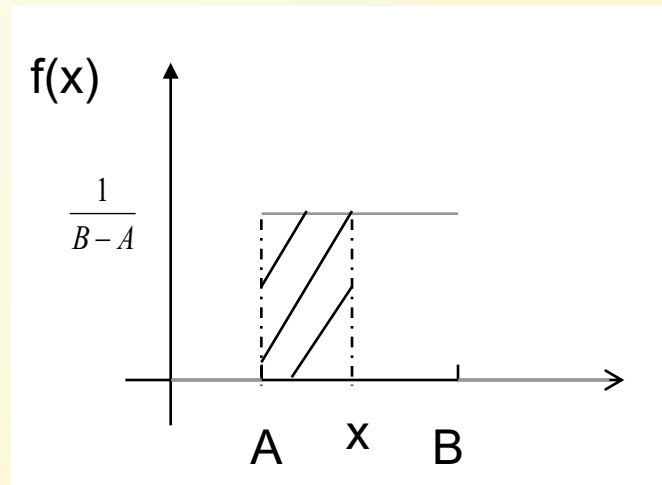


What is the cdf?

4.2 Cumulative Distribution Functions and Expected Values

■ Example 4.6 (Cont')

Solution:



For $x < A$, $F(x) = 0$, since there is no area under the graph of the density function to the left of such an x .

For $x \geq B$, $F(x) = 1$, since all the area is accumulated to the left of such an x .

4.2 Cumulative Distribution Functions and Expected Values

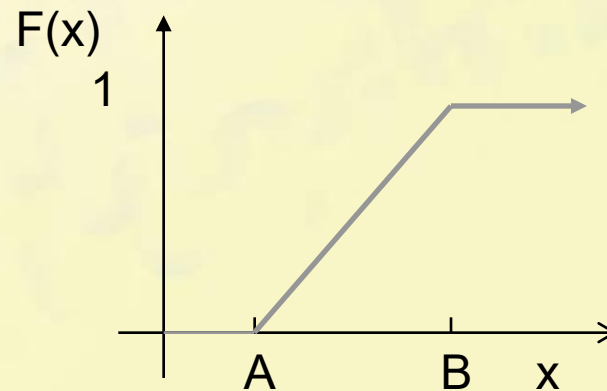
■ Example 4.6 (Cont')

For $A \leq X \leq B$

$$F(x) = \int_{-\infty}^x f(y) dy = \int_A^x \frac{1}{B-A} dy = \frac{1}{B-A} \cdot y \Big|_{y=A}^{y=x} = \frac{x-A}{B-A}$$

Therefore, the entire cdf is

$$F(x) = \begin{cases} 0 & x < A \\ \frac{x-A}{B-A} & A \leq x < B \\ 1 & x \geq B \end{cases}$$



4.2 Cumulative Distribution Functions and Expected Values

■ Using $F(x)$ to compute probabilities

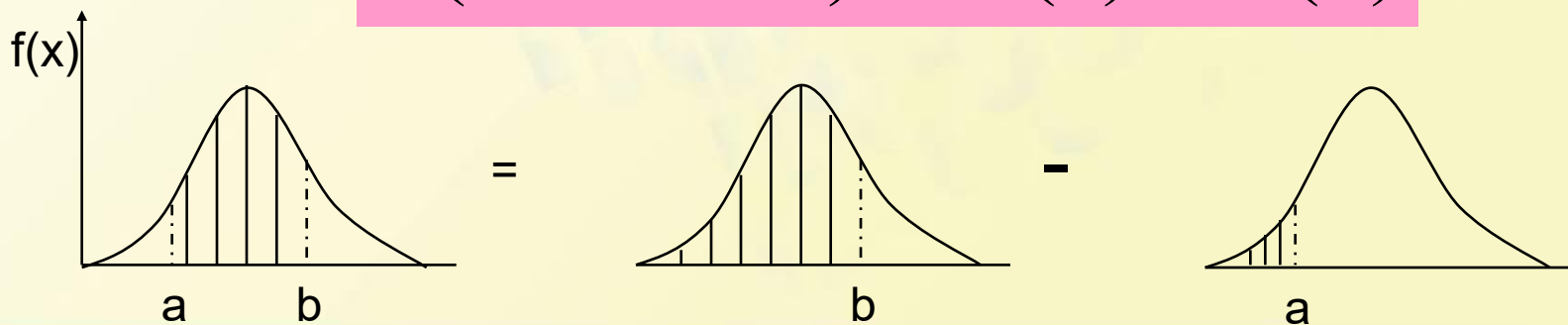
Let X be a continuous rv with pdf $f(x)$ and cdf $F(x)$.

Then for any number a

$$P(X > a) = 1 - F(a)$$

and for any two numbers a and b with $a < b$

$$P(a \leq X \leq b) = F(b) - F(a)$$



4.2 Cumulative Distribution Functions and Expected Values

■ Example 4.7

Suppose the pdf of the **magnitude X** of a **dynamic load** on a bridge is given by

$$f(x) = \begin{cases} \frac{1}{8} + \frac{3}{8}x, & 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

find $F(x)$, $P(1 \leq X \leq 1.5)$ and $P(X > 1)$?

4.2 Cumulative Distribution Functions and Expected Values

Solution:

$$F(x) = \int_{-\infty}^x f(y)dy = \int_0^x \left(\frac{1}{8} + \frac{3}{8}x\right)dy = \frac{x}{8} + \frac{3}{16}x^2$$

thus

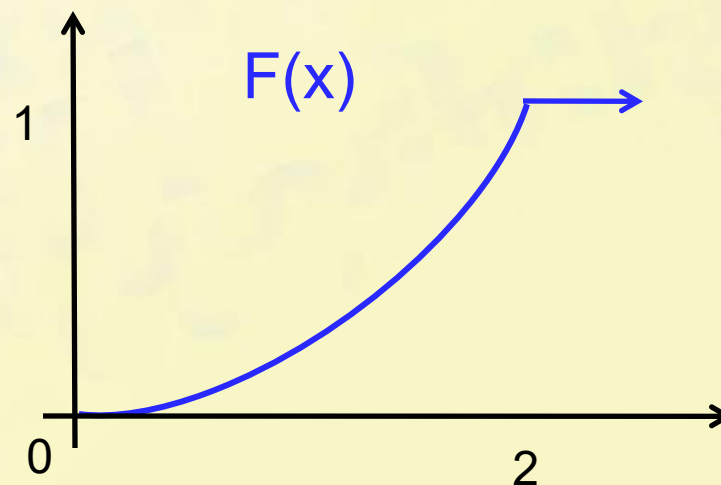
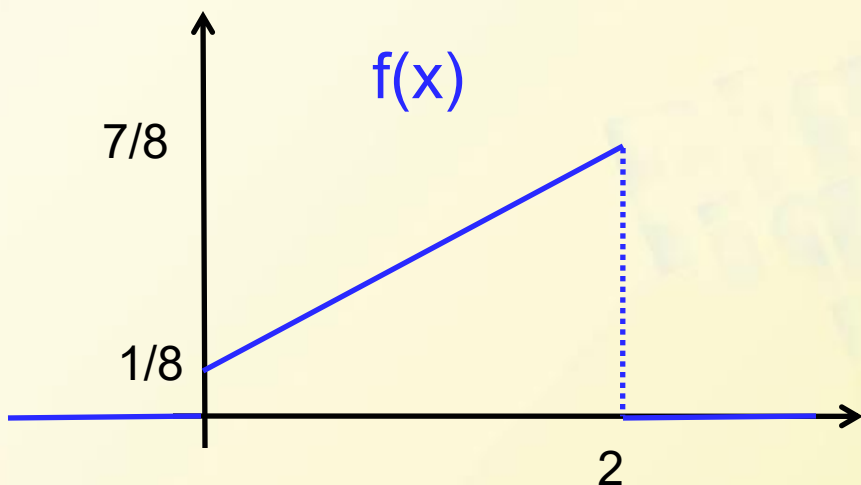
$$F(x) = \begin{cases} 0, x < 0 \\ \int_{-\infty}^x f(y)dy = \int_0^x \left(\frac{1}{8} + \frac{3}{8}x\right)dy = \frac{x}{8} + \frac{3}{16}x^2, x \in [0, 2] \\ 1, x > 2 \end{cases}$$

4.2 Cumulative Distribution Functions and Expected Values

■ Example 4.7 (Cont')

$$P(1 \leq X \leq 1.5) = F(1.5) - F(1) = 0.297$$

$$P(X > 1) = 1 - F(X = 1) = 0.688$$



4.2 Cumulative Distribution Functions and Expected Values

■ Obtaining $f(x)$ from $F(x)$

If X is a continuous rv with pdf $f(x)$ and cdf $F(x)$, then at every x at which the **derivative $F'(x)$ exists**, $F'(x)=f(x)$

$$f(x) \implies F(x) \quad F(x) = P(X \leq x) = \int_{-\infty}^x f(y)dy$$

$$F(x) \implies f(x) \quad f(x) = F'(x) = \left(\int_{-\infty}^x f(y)dy \right)'$$

4.2 Cumulative Distribution Functions and Expected Values

■ Expected/Mean Value

The expected/mean value of a **continuous rv** X with pdf $f(x)$ is

$$\mu_X = E(X) = \int_{-\infty}^{\infty} xf(x)dx$$

$$\mu_X = E(X) = \sum_{x \in D} x \cdot p(x)$$

Discrete Case

4.2 Cumulative Distribution Functions and Expected Values

■ Example 4.10 (Ex. 4.9 Cont')

The pdf of weekly gravel sales X was

$$f(x) = \begin{cases} \frac{3}{2}(1 - x^2) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

E(x)=?

Solution:

$$E(x) = \int_{-\infty}^{+\infty} xf(x)dx = \int_0^1 x \frac{3}{2}(1 - x^2)dx = \frac{3}{2} \left(\frac{x^2}{2} - \frac{x^4}{4} \right) \Big|_{x=0}^{x=1} = \frac{3}{8}$$

4.2 Cumulative Distribution Functions and Expected Values

■ Expected value of a function

If X is a **continuous rv** with pdf $f(x)$ and $h(X)$ is any function of X , then

$$E[h(X)] = \mu_{h(X)} = \int_{-\infty}^{\infty} h(x) f(x) dx$$

$$\mu_{h(X)} = E(h(X)) = \sum_{x \in D} h(x) \cdot p(x)$$

Discrete Case

4.2 Cumulative Distribution Functions and Expected Values

■ Example 4.11

Two species are competing in a region for control of a limited amount of a certain resource. Let X = the proportion of the resource controlled by species 1 and suppose X has pdf

$$f(x) = \begin{cases} 1 & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

which is a uniform distribution on $[0,1]$. Then the species that controls the majority of this resource controls the amount

$$h(X) = \max(X, 1 - X) = \begin{cases} 1 - X & \text{if } 0 \leq x < \frac{1}{2} \\ X & \text{if } \frac{1}{2} \leq X \leq 1 \end{cases}$$

What is the $E(h(X))$?

Solution:

The expected amount controlled by the species having majority control is then

$$\begin{aligned} E[h(X)] &= \int_{-\infty}^{\infty} \max(x, x-1) \cdot f(x) dx = \int_0^1 \max(x, 1-x) \cdot 1 dx \\ &= \int_0^{1/2} (1-x) \cdot 1 dx + \int_{1/2}^1 x \cdot 1 dx = \frac{3}{4} \end{aligned}$$

4.2 Cumulative Distribution Functions and Expected Values

■ The Variance

The variance of a continuous random variable X with pdf $f(x)$ and mean value μ is

$$\sigma_X^2 = V(X) = \int_{-\infty}^{+\infty} (x - \mu)^2 f(x) dx = E[(X - \mu)^2]$$

The standard deviation (SD) of X is

$$\sigma_X = \sqrt{V(X)}$$

4.2 Cumulative Distribution Functions and Expected Values

■ Proposition

$$E(aX + b) = aE(X) + b$$

$$V(X) = E(X^2) - [E(X)]^2$$

The Same Properties as Discrete Cases

4.3 The Normal Distribution

■ Normal (Gaussian) Distribution

A continuous rv X is said to have a normal distribution with parameters μ and σ (or μ and σ^2), where $-\infty < \mu < +\infty$ and $0 < \sigma$, if the pdf of X is

$$f(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)} \quad -\infty < x < \infty$$

Note:

1. The normal distribution is the most important one in all of probability and statistics. Many numerical populations have distributions that can be fit very closely by an appropriate normal curve.
2. Even when the underlying distribution is discrete, the normal curve often gives an excellent approximation.
3. Central Limit Theorem (see next Chapter)

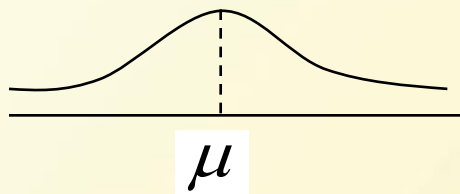
4.3 The Normal Distribution

■ Properties of $f(x;\mu,\sigma)$

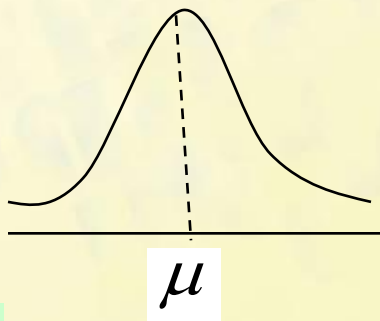
$$f(x;\mu,\sigma) \geq 0, \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)} dx = 1 \quad \text{Proof?}$$

$$E(X) = \mu \quad \& \quad V(X) = \sigma^2 \quad , \quad X \sim N(\mu, \sigma^2)$$

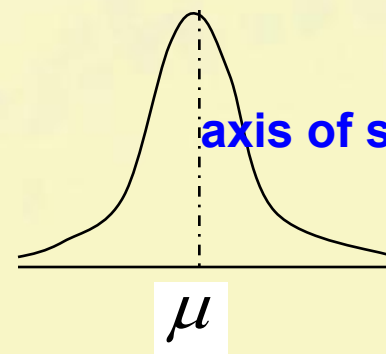
σ is large



σ is medium



σ is small



axis of symmetry

μ is location of axis of symmetry
 σ control the shape of the graph

Symmetry Shape

4.3 The Normal Distribution

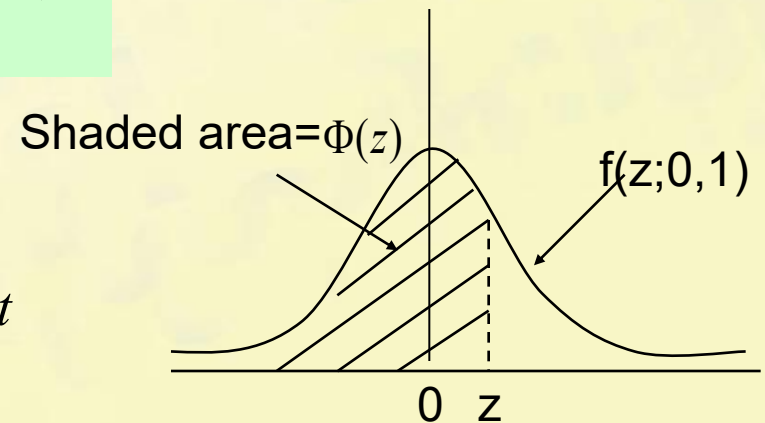
■ Standard Normal Distribution

The normal distribution with parameter values $\mu=0$ and $\sigma=1$ is called the **standard normal distribution**. A random variable that has a standard normal distribution is called a standard normal random variable and will be **denoted by Z**. **The pdf of Z is**

$$f(z;0,1) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty$$

The cdf of Z is

$$\Phi(z) = \int_{-\infty}^z f(t)dt = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$



Refer to Appendix Table A.3

4.3 The Normal Distribution

- Properties of $\Phi(z)$

$$\Phi(-z) = 1 - \Phi(z)$$

$$\Phi(0) = 0.5$$

$$P(|X| \leq z) = 2\Phi(z) - 1$$

$$P(|X| \geq z) = 2[1 - \Phi(z)]$$

Proposition

If X has the normal distribution with mean μ and standard deviation σ , then

$$Z = \frac{X - \mu}{\sigma}$$

has a standard normal distribution. Thus

$$P(a \leq X \leq b) = P\left(\frac{a - \mu}{\sigma} \leq Z \leq \frac{b - \mu}{\sigma}\right) = \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right)$$

$$P(X \leq a) = \Phi\left(\frac{a - \mu}{\sigma}\right)$$

$$P(X \geq b) = 1 - \Phi\left(\frac{b - \mu}{\sigma}\right)$$

Example

Given $X \sim N(3, 2^2)$, **find** $P(X > 2)$

Solution:

$$\text{If } X > 2, \text{ then } Y = \frac{X - \mu}{\sigma} = \frac{X - 3}{2} > -\frac{1}{2}$$

$$P(X > 2) = P(Y > -\frac{1}{2})$$

$$Y \sim N(0, 1)$$

$$= 1 - P(Y \leq -\frac{1}{2}) = 1 - \Phi(-\frac{1}{2})$$

$$= \Phi(\frac{1}{2}) = 0.6915$$

4.3 The Normal Distribution

■ Example 4.17

The breakdown voltage of a randomly chosen diode of a particular type is known to be **normally distributed**.

What is the probability that a diode's breakdown voltage is **within 1 standard deviation of its mean value**?

4.3 The Normal Distribution

Solution:

$P(X \text{ is within 1 standard deviation of its mean})$

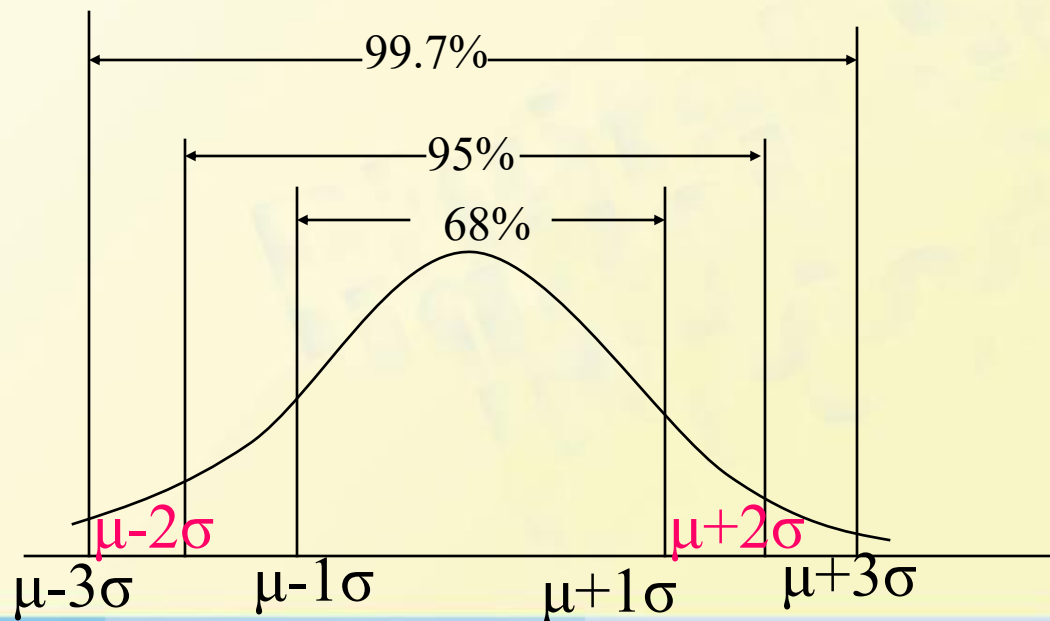
$$\begin{aligned} &= P(\mu - \sigma \leq X \leq \mu + \sigma) = P\left(\frac{\mu - \sigma - \mu}{\sigma} \leq Z \leq \frac{\mu + \sigma - \mu}{\sigma}\right) \\ &= P(-1.00 \leq Z \leq 1.00) = \Phi(1.00) - \Phi(-1.00) = 0.6826 \end{aligned}$$

Note: This question can be answered without knowing either μ or σ , as long as the distribution is known to be normal; in other words, the answer is the same for any normal distribution:

4.3 The Normal Distribution

If the population distribution of a variable is (approximately) normal, then

1. Roughly **68%** of the values are within **1 SD** of the mean.
2. Roughly **95%** of the values are within **2 SDs** of the mean
3. Roughly **99.7%** of the values are within **3 SDs** of the mean



4.3 The Normal Distribution

- The Normal Distribution and **Discrete Populations**

The **normal distribution** is often used as an approximation to the distribution of values in a **discrete population**.

4.3 The Normal Distribution

■ Proposition

Let X be a binominal rv based on n trials with success probability p . Then if the binomial probability histogram is not too skewed, X has approximately a normal distribution with $\mu = np$ and $\sigma_X = (npq)^{1/2}$.

In particular, for $x =$ a possible value of X ,

$$\begin{aligned} p(X \leq x) &= B(x; n, p) \\ &\approx (\text{area under the normal curve to the left of } x + 0.5) \\ &= \Phi\left(\frac{x + 0.5 - np}{\sqrt{npq}}\right) \end{aligned}$$

Rule: In practice, the approximation is adequate provided that both $np \geq 10$ and $nq \geq 10$. (where $q = 1 - p$)

4.3 The Normal Distribution

■ Example 4.20

Suppose that 25% of all licensed drivers in a particular state do not have insurance. **Let X be the number of uninsured drivers in a random sample of size 50, so that $p=0.25$.**

- Find:**
- (A) $P(X \leq 10)$
 - (B) $P(5 \leq X \leq 15)$

Solution:

(A) Since $np=50(0.25)=12.5 \geq 10$ and $nq=37.5 \geq 10$, **the approximation can safely be applied**. Then $\mu = 12.5$ and $\sigma = 3.06$.

$$\begin{aligned} P(X \leq 10) &= B(10; 50, 0.25) \approx \Phi\left(\frac{10 + 0.5 - 12.5}{3.06}\right) \\ &= \Phi(-0.65) = 0.2578 \end{aligned}$$

(B) Similarly, the probability that between 5 and 15 (inclusive) of the selected drivers are uninsured is

$$\begin{aligned} P(5 \leq X \leq 15) &= B(15; 50, 0.25) - B(4; 50, 0.25) \\ &\approx \Phi\left(\frac{15.5 - 12.5}{3.06}\right) - \Phi\left(\frac{4.5 - 12.5}{3.06}\right) = 0.8320 \end{aligned}$$

Chapter 5:

- Jointly Distributed Random Variables
- Expected Values, Covariance, and Correlation
- The Distribution of the Sample Mean

5.1. Jointly Distributed Random Variables

■ The Joint Probability Mass Function for Two Discrete Random Variables

Let X and Y be two discrete random variables defined on the sample space $\mathcal{S}$ of an experiment. The joint probability mass function $p(x,y)$ is defined for each pair of numbers (x,y) by

$$p(x, y) = P(X = x \text{ and } Y = y)$$

5.1. Jointly Distributed Random Variables

- Let A be any set consisting of pairs of (x,y) values. Then the probability $P[(X,Y) \in A]$ is obtained by summing the joint pmf over pairs in A :

$$p[(X,Y) \in A] = \sum_{(x,y) \in A} \sum p(x,y)$$

- Two requirements for a pmf

$$p(x,y) \geq 0 \quad \sum_x \sum_y p(x,y) = 1$$

5.1. Jointly Distributed Random Variables

■ The marginal probability mass function

The marginal probability mass functions of X and Y , denoted by $p_X(x)$ and $p_Y(y)$, respectively, are given by

$$p_X(x) = \sum_y p(x, y); \quad p_Y(y) = \sum_x p(x, y)$$

	Y_1	Y_2	$\dots$	Y_{m-1}	Y_m
X_1	$p_{1,1}$	$p_{1,2}$		$p_{1,m-1}$	$p_{1,m}$
X_2	$p_{2,1}$	$p_{2,2}$		$p_{2,m-1}$	$p_{2,m}$
$\dots$					
X_{n-1}	$p_{n-1,1}$	$p_{n-1,2}$		$p_{n-1,m-1}$	$p_{n-1,m}$
X_n	$p_{n,1}$	$p_{n,2}$		$p_{n,m-1}$	$p_{n,m}$

5.1. Jointly Distributed Random Variables

■ The Joint Probability Density Function for Two Continuous Random Variables

Let X and Y be two **continuous random variables**. Then $f(x,y)$ is the joint probability density function for X and Y if for any two-dimensional set A

$$P[(X,Y) \in A] = \iint_A f(x,y) dx dy$$

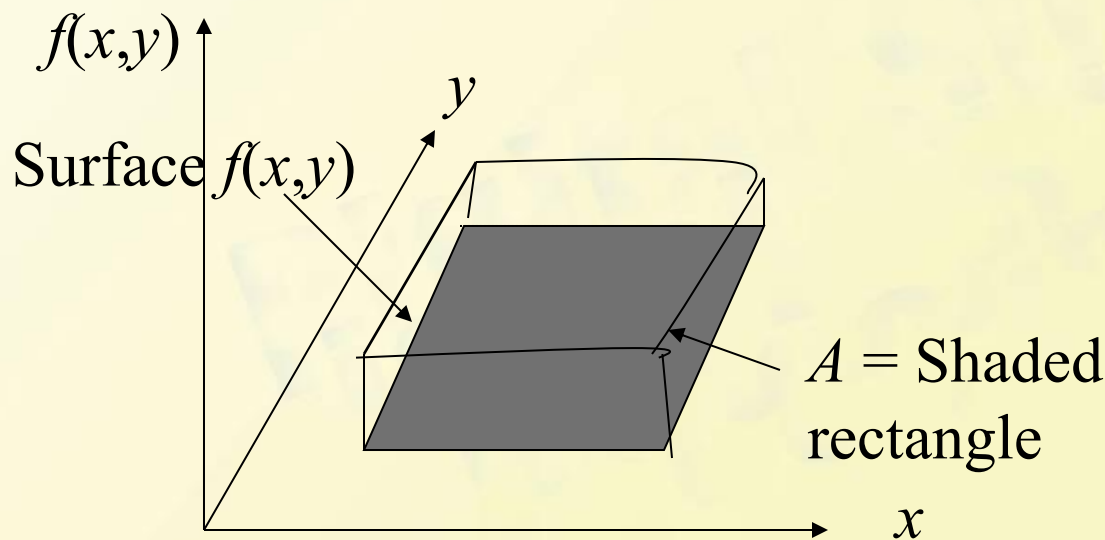
Two requirements for a joint pdf

1. $f(x,y) \geq 0$; for all pairs (x,y) in $\mathbb{R}^2$
2. $\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x,y) dx dy = 1$

5.1. Jointly Distributed Random Variables

- In particular, if A is the **two-dimensional rectangle** $\{(x,y): a \leq x \leq b, c \leq y \leq d\}$, then

$$P[(X,Y) \in A] = P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f(x,y) dy dx$$



$P[(X,Y) \in A]$ = volume under density surface above A

5.1. Jointly Distributed Random Variables

■ Example 5.3

A bank operates both a drive-up facility and a walk-up window. On a randomly selected day, let X = the proportion of time that the drive-up facility is in use, Y = the proportion of time that the walk-up window is in use. Let the joint pdf of (X, Y) be

$$f(x, y) = \begin{cases} \frac{6}{5}(x + y^2) & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

1. Verify that $f(x, y)$ is a joint probability density function;
2. Determine the probability $P(0 \leq X \leq \frac{1}{4}, 0 \leq Y \leq \frac{1}{4})$

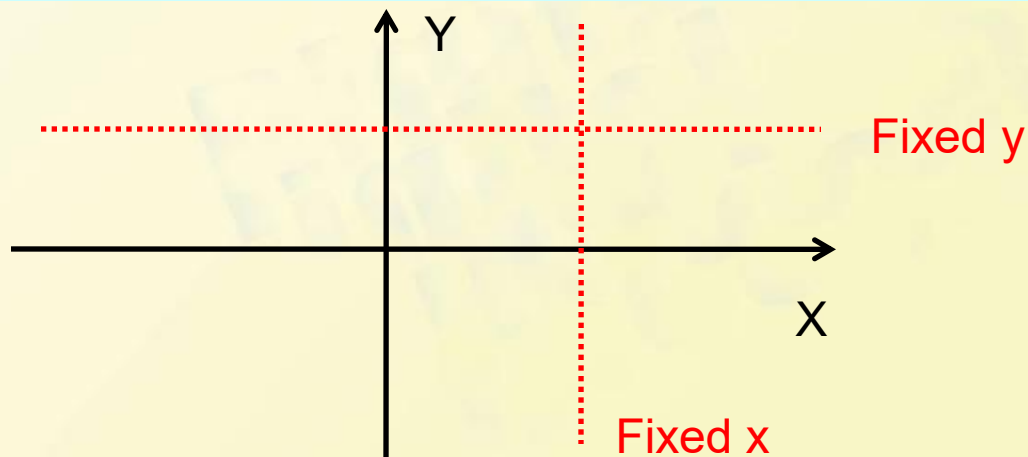
5.1. Jointly Distributed Random Variables

■ Marginal Probability density function

The marginal probability density functions of X and Y , denoted by $f_X(x)$ and $f_Y(y)$, respectively, are given by

$$f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy \quad \text{for } -\infty < x < +\infty$$

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx \quad \text{for } -\infty < y < +\infty$$



■ Example 5.5

A nut company markets cans of deluxe mixed nuts containing **almonds, cashews, and peanuts**. Suppose the net weight of each can is **exactly 1 lb**, but the weight contribution of each type of nut is random. Because the three weights sum to 1, a joint probability model for any two gives all necessary information about the weight of the third type. **Let X = the weight of almonds in a selected can and Y = the weight of cashews. The joint pdf for (X,Y) is**

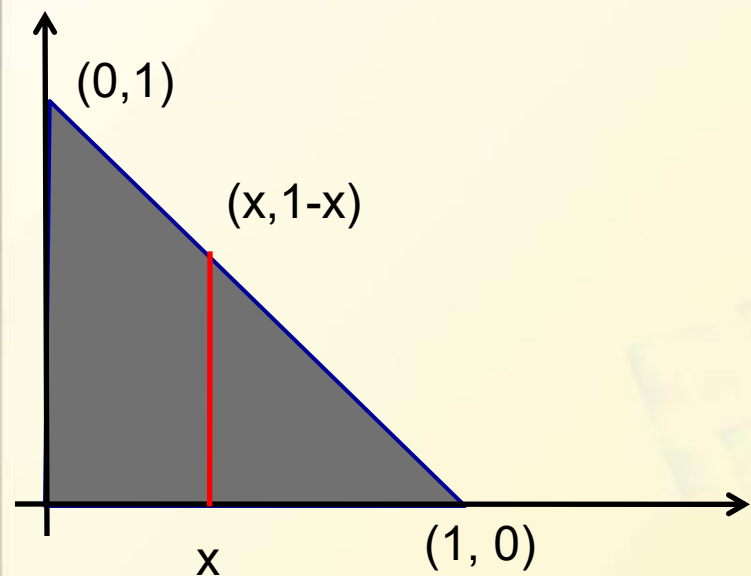
$$f(x,y) = \begin{cases} 24xy & 0 \leq x \leq 1, 0 \leq y \leq 1, x + y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

1. Verify that $f(x,y)$ is a joint probability density function;
2. Let $A = \{(x,y): 0 \leq x \leq 1, 0 \leq y \leq 1, \text{ and } x+y \leq 0.5\}$, $P(X,Y) \in A = ?$
3. $f_x(x) = ?$ And $f_y(y) = ?$

5.1. Jointly Distributed Random Variables

■ Solution:

$$f(x, y) = \begin{cases} 24xy & 0 \leq x \leq 1, 0 \leq y \leq 1, x + y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$



$$1: f(x, y) \geq 0$$

$$2: \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dy dx = \int_D \int f(x, y) dy dx$$

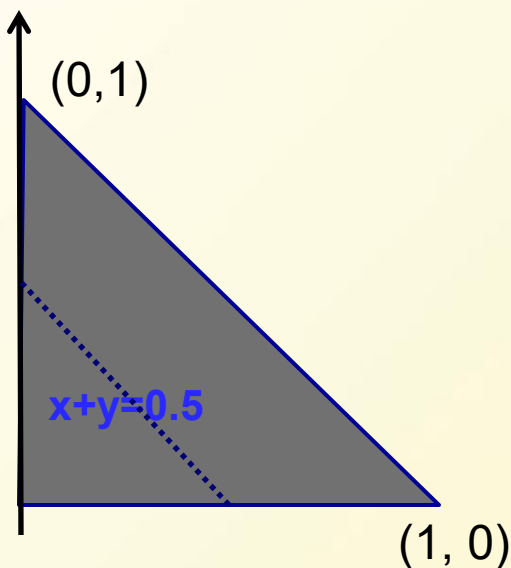
$$= \int_0^1 \left\{ \int_0^{1-x} (24xy) dy \right\} dx$$

$$= \int_0^1 12x(1-x)^2 dx = 1$$

5.1. Jointly Distributed Random Variables

■ Example 5.5 (Cont')

Let the two type of nuts together make up **at most 50%** of the can, then $A = \{(x, y); 0 \leq x \leq 1; 0 \leq y \leq 1, x + y \leq 0.5\}$

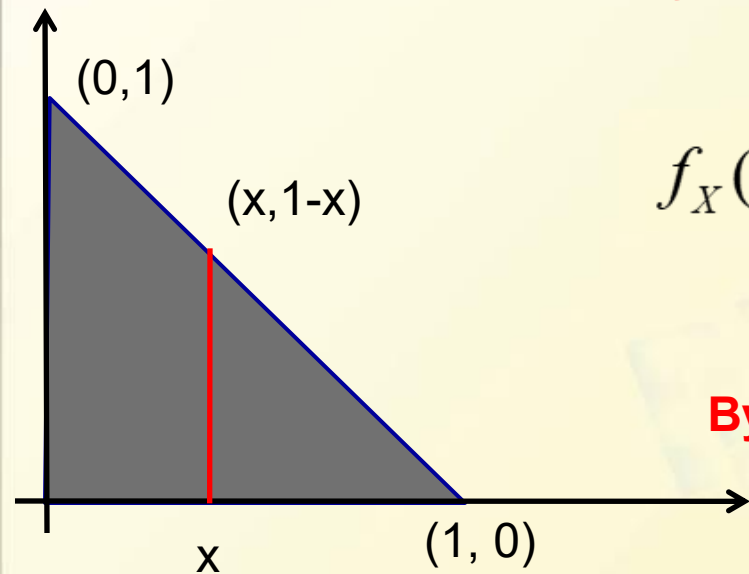


$$\begin{aligned} P((X, Y) \in A) &= \int \int_A f(x, y) dy dx \\ &= \int_0^{0.5} \left\{ \int_0^{0.5-x} (24xy) dy \right\} dx \\ &= 0.0625 \end{aligned}$$

5.1. Jointly Distributed Random Variables

■ Example 5.5 (Cont')

The **marginal pdf** for almonds is obtained by holding X fixed at x and integrating $f(x,y)$ along the vertical line through x :



$$f_X(x) = \begin{cases} \int_0^{1-x} (24xy) dy = 12x(1-x)^2, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

By symmetry

$$f_Y(y) = \int_0^{1-y} (24xy) dx = 12y(1-y)^2$$

5.1. Jointly Distributed Random Variables

■ Independent Random Variables

Two random variables X and Y are said to be independent if for **every pair of x and y values**,

$$p(x, y) = p_X(x) \cdot p_Y(y) \quad \text{when } X \text{ and } Y \text{ are discrete}$$

$$f(x, y) = f_X(x) \cdot f_Y(y) \quad \text{when } X \text{ and } Y \text{ are continuous}$$

Otherwise, X and Y are said to be dependent.

Namely, two variables are independent if their joint pmf or pdf is the product of the two marginal pmf's or pdf's.

5.1. Jointly Distributed Random Variables

■ Example 5.11

If $X_1, \dots, X_n$ represent the lifetime of **n components**, the components operate **independently of one another**, and each lifetime is **exponentially distributed** with parameter λ , then

(A) Joint pdf is?

(B) If there **n components** constitute a system that **will fail as soon as a single component fails**, then the probability that the system **lasts past t time is** ?

5.1. Jointly Distributed Random Variables

■ Solution:

$$\begin{aligned} \text{(A)} \quad f(x_1, x_2, \dots, x_n) &= (\lambda e^{-\lambda x_1})(\lambda e^{-\lambda x_2}) \dots (\lambda e^{-\lambda x_n}) \\ &= \begin{cases} \lambda^n e^{-\lambda \sum x_i}, & x_1 \geq 0; x_2 \geq 0; \dots, x_n \geq 0; \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

$$\begin{aligned} \text{(B)} \quad P(X_1 > t, X_2 > t, \dots, X_n > t) &= \int_t^\infty \dots \int_t^\infty f(x_1, x_2, \dots, x_n) dx_1 \dots dx_n \\ &= \left(\int_t^\infty \lambda e^{-\lambda x_1} dx_1 \right) \dots \left(\int_t^\infty \lambda e^{-\lambda x_n} dx_n \right) = e^{-n\lambda t} \end{aligned}$$

therefore,

$$P(\text{system lifetime} \leq t) = 1 - e^{-n\lambda t}, \text{ for } t \geq 0$$

5.2 Expected Values, Covariance, and Correlation

■ The Expected Value of a function $h(x,y)$

Let X and Y be jointly distribution rv's with **pmf** $p(x,y)$ or **pdf** $f(x,y)$ according to whether the variables are **discrete** or **continuous**. Then the expected value of a function $h(X,Y)$, denoted by $E[h(X,Y)]$ or $\mu_{h(X,Y)}$, is given by

$$E[h(X,Y)] = \begin{cases} \sum_x \sum_y h(x,y) \cdot p(x,y), & X \text{ \& } Y : \textit{discrete} \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x,y) \cdot f(x,y) dx dy, & X \text{ \& } Y : \textit{continuous} \end{cases}$$

5.2 Expected Values, Covariance, and Correlation

■ Example 5.13

Five friends have purchased tickets to a certain concert. If the tickets are for seats 1-5 in a particular row and the tickets are randomly distributed among the five, what is the expected number of seats separating any particular two of the five?

Solution:

Let X and Y denote the **seat numbers of the first and second individuals**, respectively. Possible (X, Y) pairs are $\{(1, 2), (1, 3), \dots, (5, 4)\}$, and the joint pmf of (X, Y) is $\{$

$$p(x, y) = \begin{cases} \frac{1}{20} & x = 1, \dots, 5; y = 1, \dots, 5; x \neq y \\ 0 & \text{otherwise} \end{cases}$$

The number of seats separating the two individuals is

$$h(X, Y) = |X - Y| - 1$$

■ Example 5.13 (Cont')

The accompanying table gives $h(x,y)$ for each possible (x,y) pair.

$h(x,y)$		x				
		1	2	3	4	5
y	1	--	0	1	2	3
	2	0	--	0	1	2
	3	1	0	--	0	1
	4	2	1	0	--	0
	5	3	2	1	0	--

$$\begin{aligned}
 E[h(X,Y)] &= \sum_{x=1}^5 \sum_{\substack{y=1 \\ y \neq x}}^5 h(x,y) \cdot p(x,y) \\
 &= \sum_{x=1}^5 \sum_{\substack{y=1 \\ y \neq x}}^5 (|x-y|-1) \cdot \frac{1}{20} = 1
 \end{aligned}$$

5.2 Expected Values, Covariance, and Correlation

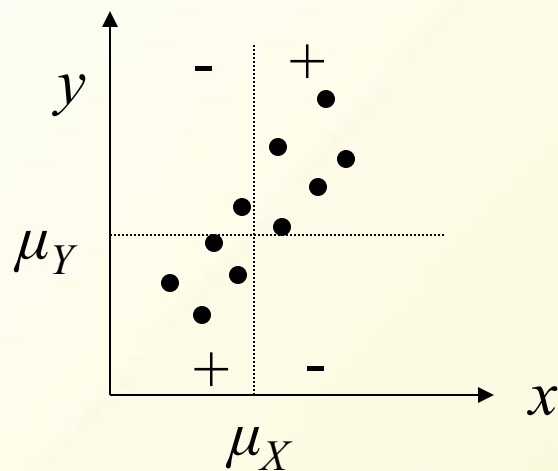
■ Covariance

When two rv X and Y are **not independent**, it is frequently of interest to assess **how strongly they are related to one another**.

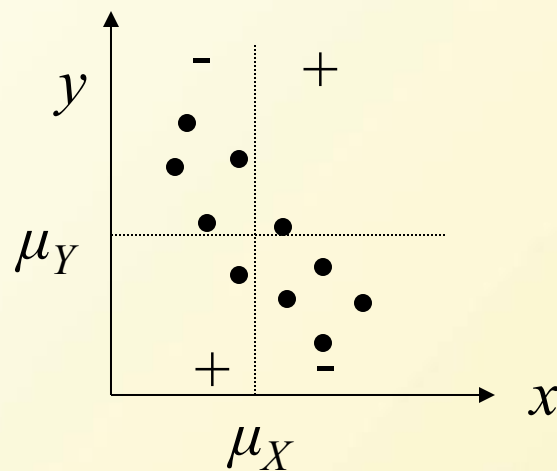
The **Covariance** between two rv's X and Y is

$$\begin{aligned}\text{Cov}(X, Y) &= E[(X - \mu_X)(Y - \mu_Y)] \\ &= \begin{cases} \sum_x \sum_y (x - \mu_X)(y - \mu_Y) p(x, y) & X, Y \text{ discrete} \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) f(x, y) dx dy & X, Y \text{ continuous} \end{cases}\end{aligned}$$

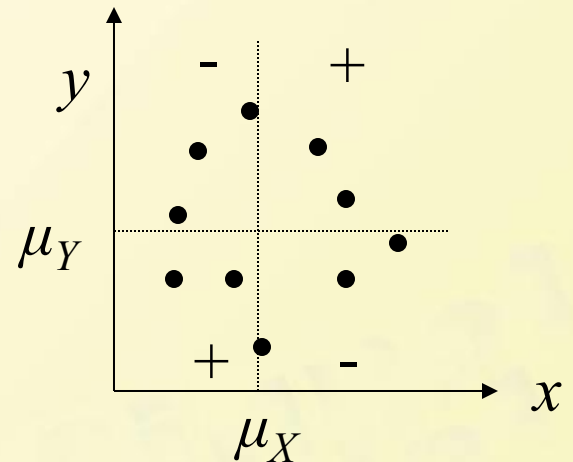
- Illustrates the different possibilities.



(a) positive covariance



(b) negative covariance;



(c) covariance near zero

Here: $P(x, y) = 1/10$

For a strong positive relation ship, $\text{Cov}(X,Y)$ should be quite positive.
For a strong negative relation ship, $\text{Cov}(X,Y)$ should be quite negative.
If X and Y are not strongly related , $\text{Cov}(X,Y)$ near 0

5.2 Expected Values, Covariance, and Correlation

■ Proposition

$$\text{Cov}(X, Y) = E(XY) - \mu_X \mu_Y$$

Note:
$$\text{Cov}(X, X) = E(X^2) - \mu_X^2 = V(X)$$

5.2 Expected Values, Covariance, and Correlation

■ Correlation

The correlation coefficient of X and Y , denoted by $\text{Corr}(X, Y)$, $\rho_{X,Y}$ or just ρ , is defined by

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \cdot \sigma_Y}$$

The normalized version of $\text{Cov}(X, Y)$

5.4 The Distribution of the Sample Mean

■ Proposition

Let $X_1, X_2, \dots, X_n$ be a **random sample** (i.i.d. rv's) from a distribution with **mean value μ and standard deviation σ** . Then

$$E(\bar{X}) = \mu_{\bar{X}} = \mu$$

$$V(\bar{X}) = \sigma_{\bar{X}}^2 = \sigma^2 / n \quad \text{and} \quad \sigma_{\bar{X}} = \sigma / \sqrt{n}$$

In addition, with **$T_o = X_1 + \dots + X_n$** (the sample total),

$$V(T_o) = n\sigma^2, \quad \text{and} \quad \sigma_{T_o} = \sqrt{n} / \sigma$$

Refer to 5.5 for the proof!

5.4 The Distribution of the Sample Mean

■ Example 5.24

In a notched tensile fatigue test on a titanium specimen, the expected number of cycles to first acoustic emission (used to indicate crack initiation) is $\mu = 28,000$, and the standard deviation of the number of cycles is $\sigma = 5000$.

Let $X_1, X_2, \dots, X_{25}$ be a random sample of size 25, where each X_i is the number of cycles on a different randomly selected specimen. Then

$$E(\bar{X}) = \mu = 28,000, E(T_0) = n\mu = 25(28000) = 700,000$$

The standard deviations of $\bar{X}$ and T_0 are

$$\sigma_{\bar{X}} = \sigma / \sqrt{n} = \frac{5000}{\sqrt{25}} = 1000$$

$$\sigma_{T_0} = \sqrt{n}\sigma = \sqrt{25}(5000) = 25,000$$

5.4 The Distribution of the Sample Mean

■ Proposition

Let $X_1, X_2, \dots, X_n$ be a random sample from a normal distribution with mean μ and standard deviation σ .

Then for **any n** , $\bar{X}$ **is normally distributed** (with mean μ and standard deviation $\sigma / \sqrt{n}$), as **is T_o** (with mean $n\mu$ and standard deviation $\sqrt{n}\sigma$).

5.4 The Distribution of the Sample Mean

■ Example 5.25

The time that it takes a randomly selected rat of a certain subspecies to find its way through a maze is a normally distributed rv with $\mu = 1.5$ min and $\sigma = .35$ min. Suppose five rats are selected. Let $X_1, X_2, \dots, X_5$ denote their times in the maze. Assuming the X_i 's to be a random sample from this normal distribution.

- **Q #1:** What is the probability that the total time $T_o = X_1 + X_2 + \dots + X_5$ for the five is between 6 and 8 min?
- **Q #2:** Determine the probability that the sample average time $\bar{X}$ is at most 2.0 min.

5.4 The Distribution of the Sample Mean

■ Example 5.25 (Cont')

A #1: T_o has a normal distribution with $\mu_{T_o} = n\mu = 5(1.5) = 7.5$ min and variance $\sigma_{T_o}^2 = n\sigma^2 = 5(0.1225) = 0.6125$, so $\sigma_{T_o} = 0.783$ min. To standardize T_o , subtract μ_{T_o} and divide by σ_{T_o} :

$$\begin{aligned} P(6 \leq T_o \leq 8) &= P\left(\frac{6-7.5}{0.783} \leq Z \leq \frac{8-7.5}{0.783}\right) \\ &= P(-1.92 \leq Z \leq 0.64) = \Phi(0.64) - \Phi(-1.92) = 0.7115 \end{aligned}$$

A #2:

$$E(\bar{X}) = \mu = 1.5 \quad \sigma_{\bar{X}} = \sigma / \sqrt{n} = 0.35 / \sqrt{5} = 0.1565$$

$$\begin{aligned} P(\bar{X} \leq 2.0) &= P\left(Z \leq \frac{2.0-1.5}{0.1565}\right) \\ &= P(Z \leq 3.19) = \Phi(3.19) = 0.9993 \end{aligned}$$

5.4 The Distribution of the Sample Mean

■ The Central Limit Theorem (CLT)

Let $X_1, X_2, \dots, X_n$ be a **random sample** from a distribution (may or may not be normal) with mean μ and variance σ^2 .

Then if n is sufficiently large, $\bar{X}$ has approximately a normal distribution with

$$\mu_{\bar{X}} = \mu, \sigma_{\bar{X}}^2 = \sigma^2 / n$$

T_o also has approximately a normal distribution with

$$\mu_{T_o} = n\mu, \sigma_{T_o}^2 = n\sigma^2$$

The larger the value of n , the better the approximation

Usually, If $n > 30$, the Central Limit Theorem can be used.

5.4 The Distribution of the Sample Mean

■ Example

When a batch of a certain chemical product is prepared, the amount of a particular impurity in the batch is a random variable **with mean value 4.0g and standard deviation 1.5g**. If 50 batches are independently prepared, what is the (approximate) probability that the sample average amount of impurity $\bar{X}$ is between 3.5 and 3.8g?

Here $n = 50$ is large enough for the CLT to be applicable. $\bar{X}$ then has approximately a normal distribution with mean value $\mu_{\bar{X}} = 4.0$ and $\sigma_{\bar{X}} = 1.5 / \sqrt{50} = 0.2121$, so

$$P(3.5 \leq \bar{X} \leq 3.8) \approx P\left(\frac{3.5 - 4.0}{0.2121} \leq Z \leq \frac{3.8 - 4.0}{0.2121}\right) = \Phi(-0.94) - \Phi(-2.36) = 0.1645$$

Chapter 6:

- Method of Moments
- Maximum likelihood Estimation

6.2 Methods of Point Estimation

■ The Method of Moment

•

The basic idea of this method is **equate** certain sample characteristics, such as the mean, **to the corresponding population** expected values. Then solving these equations for unknown parameter values yields the estimators.

6.2 Methods of Point Estimation

■ Moments

Let $X_1, X_2, \dots, X_n$ be a random sample from a pmf or pdf $f(x)$. For $k = 1, 2, 3, \dots$, the k th population moment, or k th moment of the distribution $f(x)$, is $E(X^k)$. The k th sample moment is

$$(1/n) \sum_{i=1}^n X_i^k$$

6.2 Methods of Point Estimation

■ Moment Estimator

Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with pmf or pdf $f(x; \theta_1, \dots, \theta_m)$, where $\theta_1, \dots, \theta_m$ are parameters whose values are unknown. Then the moment estimators $\hat{\theta}_1, \dots, \hat{\theta}_m$ are obtained by equating the first m sample moments to the corresponding first m population moments and solving for $\theta_1, \dots, \theta_m$.

With the given sample x_i

$$(1/n) \sum_{i=1}^n X_i^k$$

K-th sample moment

n is large

$$\Rightarrow E(X^k)$$

With unknown θ_i

K-th population moment

6.2 Methods of Point Estimation

General Algorithm :

$$\left\{ \begin{array}{l} \mu_1 = \mu_1(\theta_1, \theta_2, \dots, \theta_m) \\ \mu_2 = \mu_2(\theta_1, \theta_2, \dots, \theta_m) \\ \dots \\ \mu_m = \mu_m(\theta_1, \theta_2, \dots, \theta_m) \end{array} \right. \quad \Rightarrow \quad \left\{ \begin{array}{l} \theta_1 = \theta_1(\mu_1, \mu_2, \dots, \mu_m) \\ \theta_2 = \theta_2(\mu_1, \mu_2, \dots, \mu_m) \\ \dots \\ \theta_m = \theta_m(\mu_1, \mu_2, \dots, \mu_m) \end{array} \right.$$

The first m population moments

The solution of equations

Use the **first m sample moment**

$$A_l = \frac{1}{n} \sum_{i=1}^n X_i^l, l = 1, 2, \dots, m$$

to represent the **population moments μ_i**

$$\theta_i = \theta_i(A_1, A_2, \dots, A_m), i = 1, \dots, m$$

6.2 Methods of Point Estimation

■ Example 6.12

Let $X_1, X_2, \dots, X_n$ represent a random sample of service times of n customers at a certain facility, where the underlying distribution is assumed exponential with parameter λ . How to estimate λ by using the method of moments?

Step #1: The 1st population moment $E(X) = 1/\lambda$
then we have $\lambda = 1 / E(X)$

Step #2: Use the 1st sample moment $\bar{X}$ to represent 1st population moment $E(X)$, and get the estimator

$$\hat{\lambda} = 1 / \bar{X}$$

6.2 Methods of Point Estimation

■ Example 6.13

Let $X_1, \dots, X_n$ be a random sample from a gamma distribution with parameters α and β . Its pdf is

$$f(x; \alpha, \beta) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-\frac{x}{\beta}} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

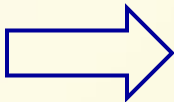
There **are two parameter** need to be estimated, thus, consider the **first two moments**

6.2 Methods of Point Estimation

■ Example 6.13 (Cont')

Step #1: $E(X) = \mu = \alpha\beta$

$$E(X^2) = V(X) + [E(X)]^2 = \alpha\beta^2 + \alpha^2\beta^2 = \alpha\beta^2(1 + \alpha)$$

 $\alpha = \frac{E(X)^2}{E(X^2) - E(X)^2}, \beta = \frac{E(X^2) - E(X)^2}{E(X)}$

Step #2:

$$\bar{X} \rightarrow E(X), \frac{1}{n} \sum X_i^2 \rightarrow E(X^2)$$

$$\hat{\alpha} = \frac{\bar{X}^2}{\frac{1}{n} \sum X_i^2 - \bar{X}^2} \quad \hat{\beta} = \frac{\frac{1}{n} \sum X_i^2 - \bar{X}^2}{\bar{X}}$$

6.2 Methods of Point Estimation

■ Example 6.14

Let $X_1, \dots, X_n$ be a random sample from a generalized **negative binomial distribution** with parameters r and p . Its pmf is

$$nb(x; r, p) = \binom{x + r - 1}{r - 1} p^r (1 - p)^x, \quad x = 0, 1, 2, \dots$$

Determine the moment estimators of parameters r and p .

Note: There are two parameters needs to estimate, thus the first two moments are considered.

6.2 Methods of Point Estimation

■ Example 6.14 (Cont')

Step #1: $E(X) = r(1-p) / p$

$$E(X^2) = r(1-p)(r-rp+1) / p^2$$

⇒
$$p = \frac{E(X)}{E(X^2) - E(X)^2}, r = \frac{E(X)^2}{E(X^2) - E(X)^2 - E(X)}$$

Step #2: $\bar{X} \rightarrow E(X), \frac{1}{n} \sum X_i^2 \rightarrow E(X^2)$

$$\hat{p} = \frac{\bar{X}}{\frac{1}{n} \sum X_i^2 - \bar{X}^2} \quad \hat{r} = \frac{\bar{X}^2}{\frac{1}{n} \sum X_i^2 - \bar{X}^2 - \bar{X}}$$

6.2 Methods of Point Estimation

■ Maximum Likelihood Estimation

Let $X_1, X_2, \dots, X_n$ have joint pmf or pdf

$$f(x_1, x_2, \dots, x_n; \theta_1, \dots, \theta_m)$$

where the parameters $\theta_1, \dots, \theta_m$ have unknown values. When $x_1, \dots, x_n$ are the observed sample values and f is regarded as a function of $\theta_1, \dots, \theta_m$, it is called the likelihood function.

The maximum likelihood estimates(mle's) $\hat{\theta}_1, \dots, \hat{\theta}_m$ are those values of the θ_i 's that maximize the likelihood function, so that

$$f(x_1, \dots, x_n; \hat{\theta}_1, \dots, \hat{\theta}_m) \geq f(x_1, \dots, x_n; \theta_1, \dots, \theta_m) \quad \text{for all } \theta_1, \dots, \theta_m$$

When the X_i 's are substituted in place of the x_i 's, the maximum likelihood estimators result.

6.2 Methods of Point Estimation

■ Three steps

1. Write the joint pmf/pdf (i.e. Likelihood function)

$$f(x_1, x_2, \dots, x_n; \theta_1, \dots, \theta_m) = \prod_{i=1}^n f(x_i; \theta_1, \dots, \theta_m)$$

2. Get the $\ln(\text{likelihood})$ **(if necessary)**

$$\ln[f(x_1, x_2, \dots, x_n; \theta_1, \dots, \theta_m)] = \sum_{i=1}^n \ln(f(x_i; \theta_1, \dots, \theta_m))$$

3. Take the partial derivative of $\ln(f)$ with respect to θ_i , **equal them to 0**, and solve the resulting m equations.

$$\frac{d}{d\theta_i} \ln[f(x_1, x_2, \dots, x_n; \theta_1, \dots, \theta_m)] = 0$$

6.2 Methods of Point Estimation

■ Example 6.16

Suppose $X_1, X_2, \dots, X_n$ is a random sample from an **exponential distribution** with the unknown parameter λ . Determine the **maximum likelihood estimator of λ** .

The joint pdf is (independence)

$$f(x_1, \dots, x_n; \lambda) = (\lambda e^{-\lambda x_1}) \cdots (\lambda e^{-\lambda x_n}) = \lambda^n e^{-\lambda \sum x_i}$$

The $\ln(\text{likelihood})$ is $\ln[f(x_1, \dots, x_n; \lambda)] = n \ln(\lambda) - \lambda \sum x_i$

Equating to zero the derivative w.r.t. λ :

$$\frac{d \ln[f(x_1, \dots, x_n; \lambda)]}{d \lambda} = \frac{n}{\lambda} - \sum x_i = 0 \quad \Rightarrow \quad \lambda = \frac{n}{\sum x_i} = \frac{1}{\bar{x}} \quad \Rightarrow \quad \hat{\lambda} = 1 / \bar{X} \quad \text{The estimator}$$

6.2 Methods of Point Estimation

■ Example 6.17

Let $X_1, X_2, \dots, X_n$ is a random sample from **a normal distribution** $N(\mu, \sigma^2)$. Determine the maximum likelihood estimator of μ and σ^2 .

The joint pdf is

$$f(x_1, \dots, x_n; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_1-\mu)^2}{2\sigma^2}} \cdot \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_2-\mu)^2}{2\sigma^2}} \dots \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_n-\mu)^2}{2\sigma^2}} = \left(\frac{1}{2\pi\sigma^2} \right)^{n/2} e^{-\frac{\sum (x_i-\mu)^2}{2\sigma^2}}$$

$$\ln[f(x_1, \dots, x_n; \mu, \sigma^2)] = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (x_i - \mu)^2$$

Equating to 0 the partial derivatives w.r.t. μ and σ^2 , finally we have

$$\hat{\mu} = \bar{X}, \quad \sigma^2 = \frac{\sum (X_i - \bar{X})^2}{n}$$

Here the mle of σ^2 is not the unbiased estimator.

Chapter 7:

- 7.1. Basic Properties of Confidence Intervals
- 7.2. Larger-Sample Confidence Intervals for a Population Mean and Proportion
- 7.3 Intervals Based on a Normal Population Distribution

7.1 Basic Properties of Confidence Intervals

■ Considering a Simple Case

Suppose that the parameter of interest is a **population mean μ** and that

1. The population distribution **is normal**.
2. The value of the population standard deviation **σ is known**

- Normality of the population distribution is often a reasonable assumption.
- If the value of μ is unknown, it is implausible that the value of σ would be available.

In later sections, we will develop methods based on less restrictive assumptions.

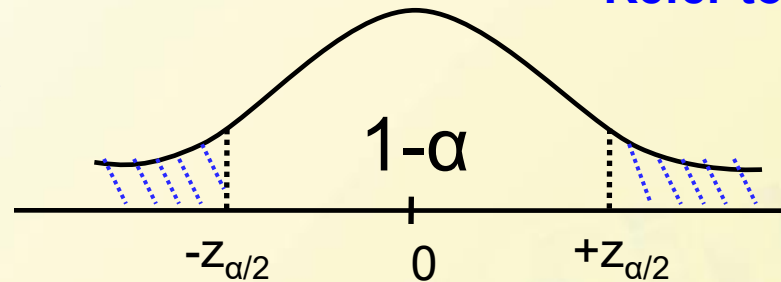
7.1 Basic Properties of Confidence Intervals

■ Other Levels of Confidence

$$P(a < z < b) = 1 - \alpha$$

Why is Symmetry?
Refer to pp. 276 Ex.8

Refer to pp.156 for the Definition Z_α



A $100(1 - \alpha)\%$ confidence interval for the mean μ of a normal population when the value of σ is known is given by

$$\left(\bar{x} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \right) \text{ or, } \bar{x} \pm z_{\alpha/2} \cdot \sigma / \sqrt{n}$$

For instance, the 99% CI is $\bar{x} \pm 2.58 \cdot \sigma / \sqrt{n}$

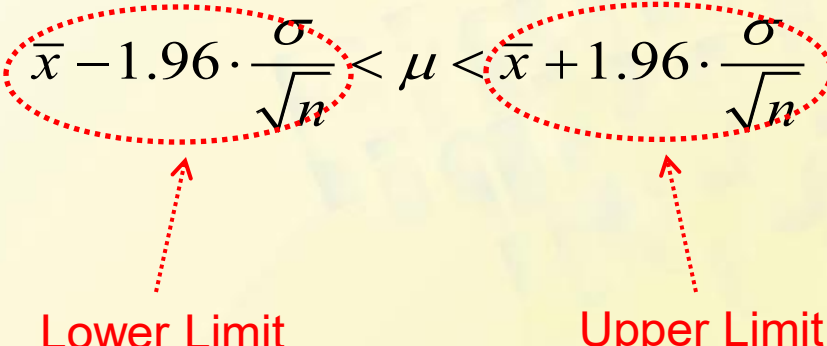
7.1 Basic Properties of Confidence Intervals

■ Definition

If after observing $X_1=x_1, X_2=x_2, \dots, X_n=x_n$, we compute the observed **sample mean** $\bar{x}$. The resulting fixed interval is called **a 95% confidence** interval for μ . This CI can be expressed either as

$$\left(\bar{x} - 1.96 \cdot \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \cdot \frac{\sigma}{\sqrt{n}} \right) \quad \text{is a 95\% CI for } \mu$$

or as $\bar{x} - 1.96 \cdot \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + 1.96 \cdot \frac{\sigma}{\sqrt{n}}$ with a 95% confidence



Lower Limit

Upper Limit

7.1 Basic Properties of Confidence Intervals

■ *Example 7.3*

Let's calculate a confidence interval for true average hole diameter using **a confidence level of 90%**.

This requires that $100(1-\alpha) = 90$, from which $\alpha = 0.1$ and $z_{\alpha/2} = z_{0.05} = 1.645$. The desired interval is then

$$5.426 \pm (1.645) \cdot \frac{0.100}{\sqrt{40}} = 5.426 \pm 0.26 = (5.400, 5.452)$$

7.1 Basic Properties of Confidence Intervals

■ Confidence Level, Precision, and Choice of Sample Size

$$\left(\bar{x} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \right)$$

Then the width (Precision) of the CI

$$w = 2 \times z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

Independent of the sample mean

Higher confidence level (larger $z_{\alpha/2}$) → A wider interval

Reliability ↔ **Precision**

Larger σ → A wider interval

Smaller n → A wider interval

Given a desired confidence level (α) and interval width (w), then we can determine the necessary **sample size n** , by

$$n = \left(2z_{\alpha/2} \cdot \frac{\sigma}{w} \right)^2$$

7.2 Large-Sample Confidence Intervals for a Population Mean and Proportion

- The CI for μ given in the previous section assumed that the **population distribution is normal** and that the value of **σ is known**. We now present a **large-sample** CI whose validity **does not require these assumptions**.
- Let $X_1, X_2, \dots, X_n$ be a random sample from a population having a mean μ and standard deviation σ (**any population, normal or un-normal**).
Provided that n is large (**Large-Sample**), the Central Limit Theorem (CLT) implies that $\bar{X}$ **has approximately a normal distribution** whatever the nature of the population distribution.

7.2 Large-Sample Confidence Intervals for a Population Mean and Proportion

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \sim N(0,1) \Rightarrow P(-z_{\alpha/2} < \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} < z_{\alpha/2}) \approx 1 - \alpha$$

Therefore, $\bar{x} \pm z_{\alpha/2} \cdot \sigma / \sqrt{n}$ is a large-sample CI for μ with a confidence level of **approximately** $100(1 - \alpha)\%$.

That is , **when n is large, the CI for μ given previously remains valid whatever the population distribution, provided that the qualifier “approximately” is inserted in front of the confidence level.**

When σ is not known, which is generally the case, we may consider the following standardized variable

$$Z = \frac{\bar{X} - \mu}{S / \sqrt{n}} \quad S \approx \sigma$$

7.2 Large-Sample Confidence Intervals for a Population Mean and Proportion

■ Proposition

If n is sufficiently large (**usually, $n > 40$**), the standardized variable

$$Z = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has approximately a standard normal distribution, meaning that

$$\bar{x} \pm z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

Compared with (7.5) in pp.272

$$s \approx \delta$$

is a large-sample confidence interval for μ with confidence level **approximately** $100(1-\alpha)\%$.

Note: This formula is valid regardless of the shape of the population distribution.

7.2 Large-Sample Confidence Intervals for a Population Mean and Proportion

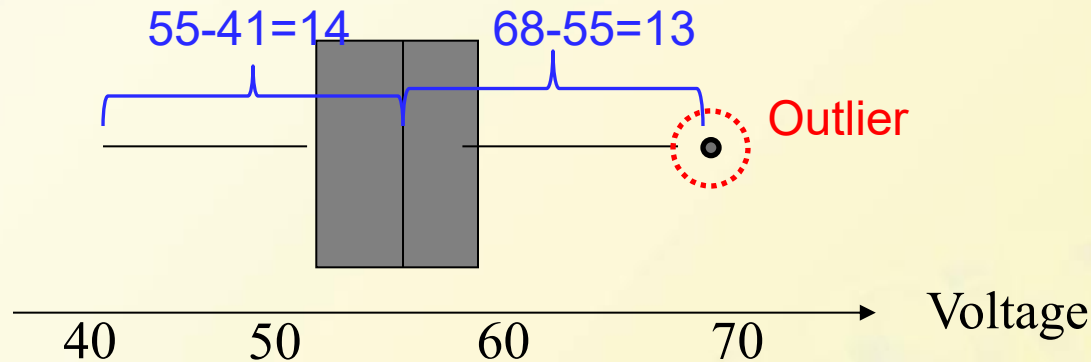
■ Example 7.6

The alternating-current breakdown voltage of an insulating liquid indicates its dielectric strength. The article “test practices for the AC breakdown voltage testing of insulation liquids,” gave the accompanying sample observations on breakdown voltage of a particular circuit under certain conditions.

62 50 53 57 41 53 55 61 59 64 50 53 64 62 50 68
54 55 57 50 55 50 56 55 46 55 53 54 52 47 47 55
57 48 63 57 57 55 53 59 53 52 50 55 60 50 56 58

7.2 Large-Sample Confidence Intervals for a Population Mean and Proportion

■ Example 7.6 (Cont')



Summary quantities include

$$n = 48, \sum x_i = 2626, \sum x_i^2 = 144950 \Rightarrow \bar{x} = 54.7 \text{ and } s = 5.23$$

The 95% confidence interval is then

$$54.7 \pm 1.96 \frac{5.23}{\sqrt{48}} = 54.7 \pm 1.5 = (53.2, 56.2)$$

7.3 Intervals Based on a Normal Population Distribution

- The CI for μ presented in the previous section is valid provided **that n is large**. The resulting interval can be used **whatever** the nature of the population distribution (with unknown μ and σ).
- **If n is small, the CLT can not be invoked. In this case we should make a specific assumption.**
- Assumption

The population of interest is normal, $X_1, X_2, \dots, X_n$ constitutes a random sample from **a normal distribution** with both **μ and σ unknown**.

7.3 Intervals Based on a Normal Population Distribution

■ Theorem

When $\bar{X}$ is the mean of a random sample of size n from a normal distribution with mean μ . Then the rv

$$T = \frac{\bar{X} - \mu}{S / \sqrt{n}}$$

has a probability distribution called a **t distribution** with **$n-1$** degrees of freedom (df) .

only $n-1$ of these are “freely determined”

S is based on the **n** deviations $(X_1 - \bar{X}), (X_2 - \bar{X}), \dots, (X_n - \bar{X})$

Notice that $\sum_{i=1}^n (X_i - \bar{X}) = 0$

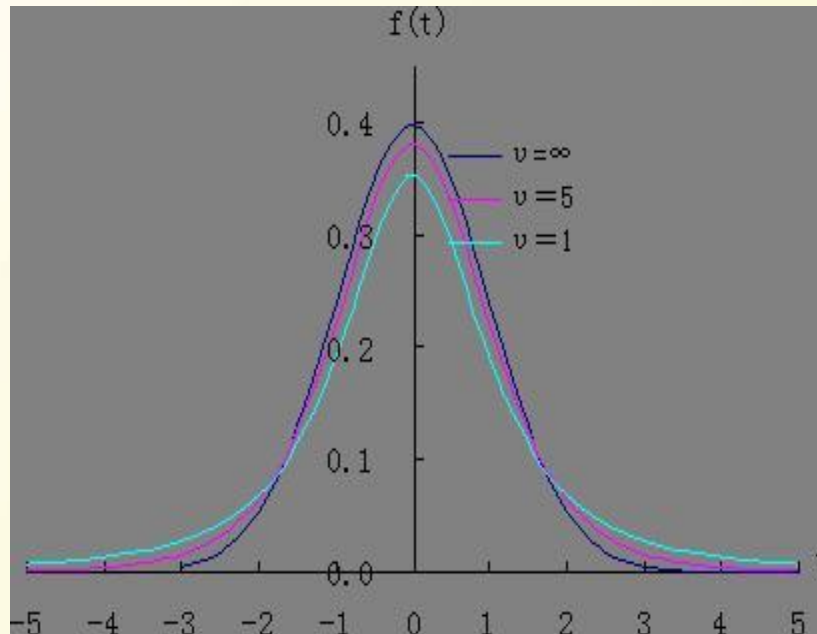
7.3 Intervals Based on a Normal Population Distribution

■ Properties of t Distributions

$$T = \frac{\bar{X} - \mu}{S / \sqrt{n}}$$

The only one parameter in T is the number of df: $v=n-1$

Let t_v be the density function curve for v df



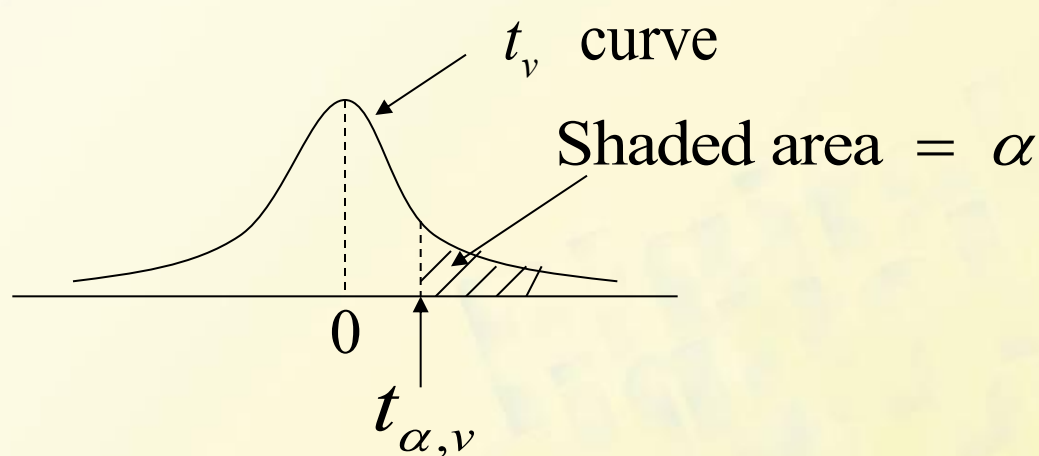
1. Each t_v curve is bell-shaped and centered at 0.
2. Each t_v curve is more spread out than the standard normal curve.
3. As v increases, the spread of the corresponding t_v curve decreases.
4. As $v \rightarrow \infty$, the sequence of t_v curves approaches the standard normal curve $N(0,1)$.

Rule: $v \geq 40 \sim N(0,1)$

7.3 Intervals Based on a Normal Population Distribution

■ Notation

Let $t_{\alpha, v}$ = the value on the measurement axis for which the area under the t curve with v df to the right of $t_{\alpha, v}$ is α ; $t_{\alpha, v}$ is called a **t critical value**



Fixed α , $v \uparrow$, $t_{\alpha, v} \downarrow$

Fixed v , $\alpha \uparrow$, $t_{\alpha, v} \downarrow$

Figure 7.7 A pictorial definition of $t_{\alpha, v}$ (Refer to: Appendix Table A.5)

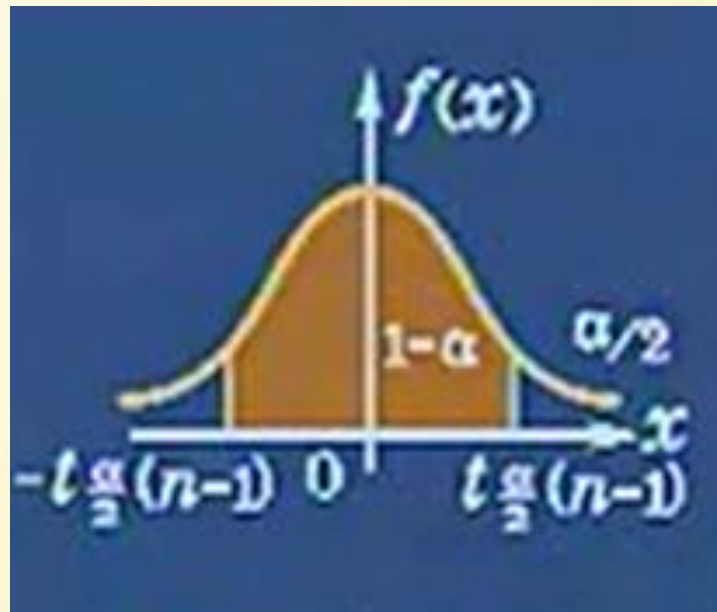
Refer to pp.156 for the similar definition of Z_α

7.3 Intervals Based on a Normal Population Distribution

■ The One-Sample t confidence Interval

The standardized variable T has a t distribution with $n-1$ df, and the area under the corresponding t density curve between $-t_{\alpha/2, n-1}$ and $t_{\alpha/2, n-1}$ is $1-\alpha$, so

$$P(-t_{\alpha/2, n-1} < T < t_{\alpha/2, n-1}) = 1 - \alpha$$



7.3 Intervals Based on a Normal Population Distribution

Proposition

Let $\bar{x}$ and s be the **sample mean** and **sample standard deviation** computed from the results of a random sample from **a normal population** with mean μ . Then a $100(1-\alpha)\%$ confidence interval for μ is

$$\left(\bar{x} - t_{\alpha/2, n-1} \cdot \frac{s}{\sqrt{n}}, \bar{x} + t_{\alpha/2, n-1} \cdot \frac{s}{\sqrt{n}} \right) \quad \text{Or, compactly} \quad \bar{x} \pm t_{\alpha/2, n-1} \cdot s / \sqrt{n}$$

An upper confidence bound with $100(1-\alpha)\%$ confidence level for μ is $\bar{x} + t_{\alpha, n-1} \cdot s / \sqrt{n}$.
Replacing $+$ by $-$ gives a lower confidence bound for μ .

7.3 Intervals Based on a Normal Population Distribution

■ Example 7.11

Consider the following observations

10490 16620 17300 15480 12970 17260 13400 13900
13630 13260 14370 11700 15470 17840 14070 14760

1. Approximately normal by observing the probability plot.
2. $n = 16$ is small, and the population deviation σ is unknown, so we choose the statistic T with a t distribution of $n - 1 = 15$ df. The resulting **95% CI** is

$$\bar{x} \pm t_{.025,15} \cdot \frac{s}{\sqrt{n}} = 14,532.5 \pm (2.131) \frac{2055.67}{\sqrt{16}} = (13,437.3, 15,627.7)$$

7.3 Intervals Based on a Normal Population Distribution

■ Example 7.12

Consider the following sample of fat content (in percentage) of $n = 10$ randomly selected hot dogs

25.2 21.3 22.8 17.0 29.8 21.0 25.5 16.0 20.9 19.5

Assume that these were selected from a normal population distribution.

Please give a 95% **CI** for the population mean fat content.

$$\begin{aligned}\bar{x} \pm t_{.025,9} \cdot \frac{s}{\sqrt{n}} &= 21.90 \pm 2.262 \cdot \frac{4.134}{\sqrt{10}} = 21.90 \pm 2.96 \\ &= (18.94, 24.86)\end{aligned}$$

Summary of Chapter 7

- General method for deriving CIs (2 properties, p.273)

Case #1: (7.1)

CI for μ of a normal distribution with known σ ;

Case #2: (7.2)

Large-sample CIs for μ of General distributions with unknown σ

Case #3: (7.3)

Small-sample CIs for μ of Gaussian distributions with unknown σ

- Both Sided Vs. One-sided CIs (p.283)